US012399135B2

(12) **United States Patent**
Mo et al.

(10) **Patent No.: US 12,399,135 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **OPTICAL INSPECTION TOOL AND INSPECTION METHOD FOR INSPECTING MULTIFACETED GROOVE SPECIMENS**

(71) Applicant: **Intel Corporation**, Santa Clara, CA (US)

(72) Inventors: **Jianyong Mo**, Chandler, AZ (US); **Fan Fan**, Chandler, AZ (US); **Liang Zhang**, Chandler, AZ (US)

(73) Assignee: **Intel Corporation**, Santa Clara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 477 days.

(21) Appl. No.: **17/737,045**

(22) Filed: **May 5, 2022**

(65) **Prior Publication Data**

US 2023/0358690 A1 Nov. 9, 2023

(51) **Int. Cl.**
G01N 21/956 (2006.01)
G01N 21/95 (2006.01)
H04N 23/90 (2023.01)

(52) **U.S. Cl.**
CPC ***G01N 21/956*** (2013.01); ***G01N 21/9501*** (2013.01); ***H04N 23/90*** (2023.01)

(58) **Field of Classification Search**
CPC G01N 21/956; G01N 21/9501; G01N 21/9515; H04N 23/90
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,812,614 A *	3/1989	Wang	G05B 19/425	901/42
4,989,338 A *	2/1991	Tsuji	G01B 21/20	33/833
7,751,046 B2 *	7/2010	Levy	G01N 21/27	356/636
9,922,419 B2 *	3/2018	Takahashi	B65B 37/04	
10,895,450 B2 *	1/2021	Lau	G06T 7/593	
11,604,062 B2 *	3/2023	Haugen	G01B 11/25	
2007/0023716 A1 *	2/2007	van der Burgt	G06T 7/001	250/559.29
2010/0208980 A1 *	8/2010	Urban	G01N 21/9501	382/149
2016/0104282 A1 *	4/2016	Takahashi	H04N 23/10	382/103
2019/0128666 A1 *	5/2019	Lau	G01B 11/2545	
2020/0124407 A1 *	4/2020	Haugen	G06T 7/521	
2020/0124410 A1 *	4/2020	Haugen	H04N 25/71	

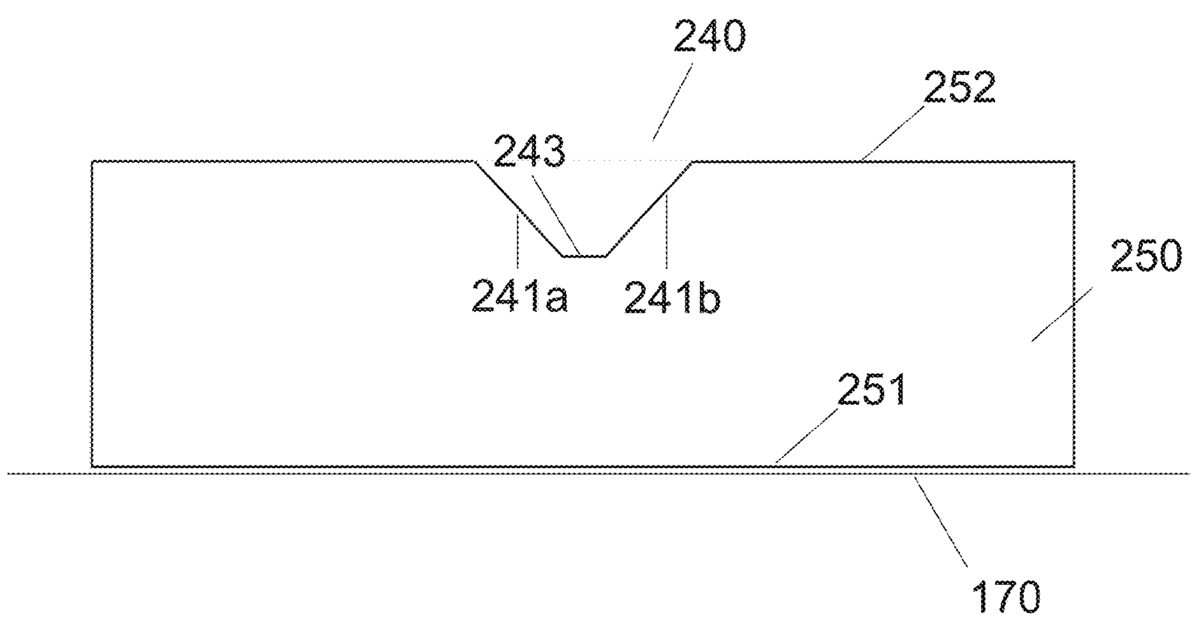
* cited by examiner

Primary Examiner — Roy M Punnoose
(74) *Attorney, Agent, or Firm* — Viering, Jentschura & Partner mbB

(57) **ABSTRACT**

An optical inspection tool may include at least a first image capture unit and a second image capture unit for inspecting specimens having a substantially V-shaped grooves. The first image capture unit may be arranged in a first orientation so as to be directable towards a first angular surface of the V-shaped groove of each specimen. The second image capture unit may be arranged in a second orientation so as to be directable towards a second angular surface of the V-shaped groove of each specimen. The first image capture unit may be configured to capture images of defects and/or contamination on the first angular surface and the second image capture unit may be configured to capture images of defects and/or contamination on the second angular surface.

20 Claims, 15 Drawing Sheets



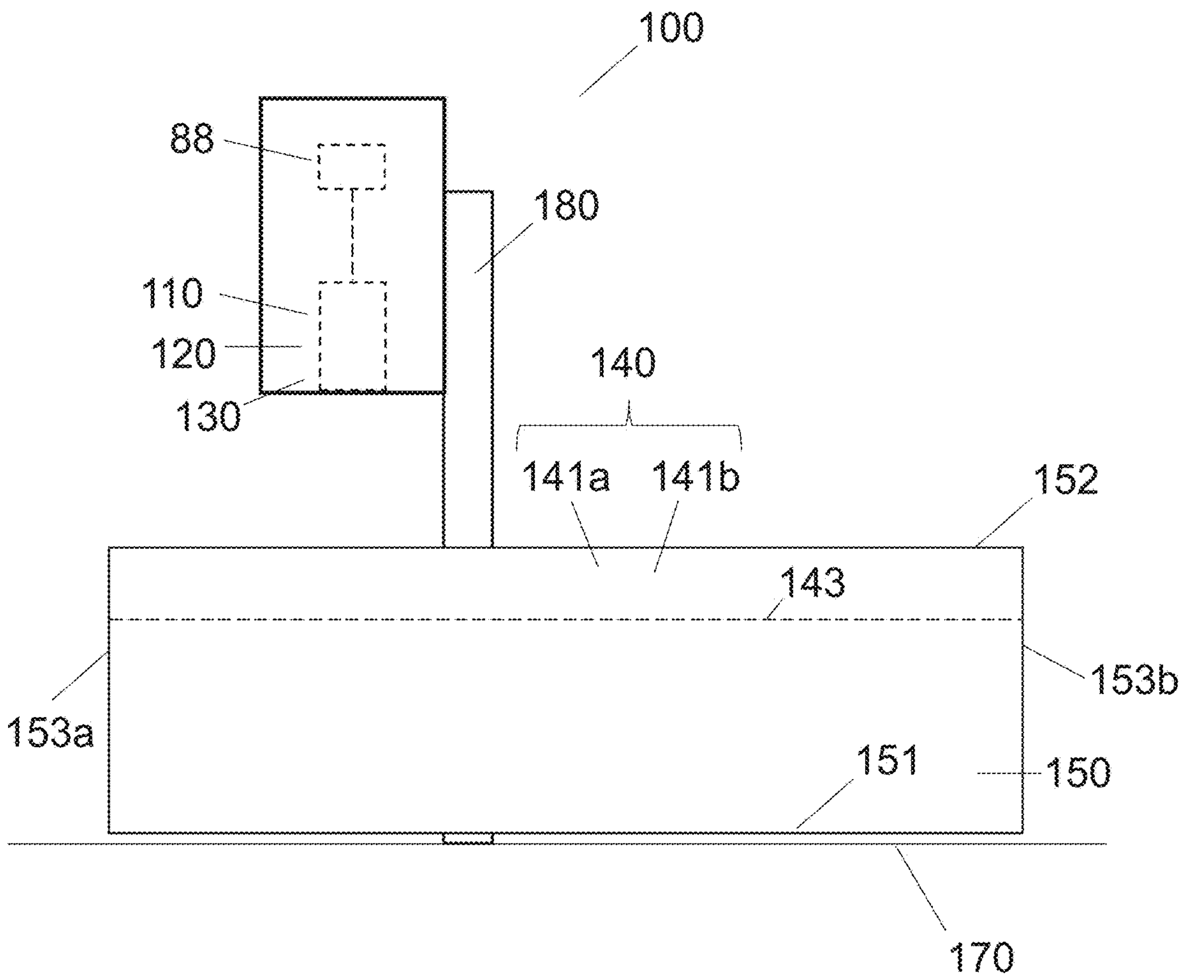

100
88
180
110
120
130
140
141a
141b
152
143
153a
153b
151
150
170

FIG. 1

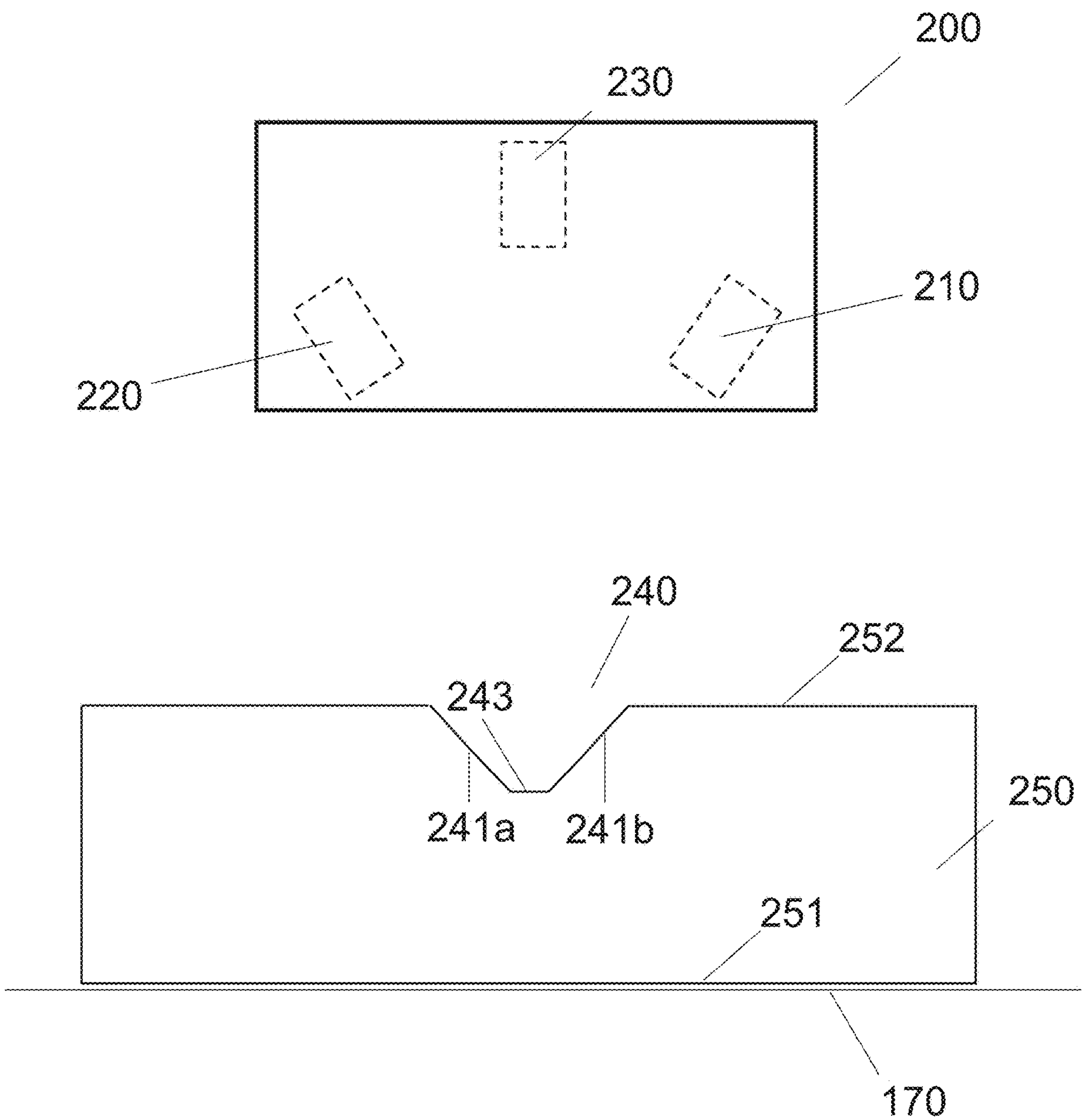


FIG. 2A

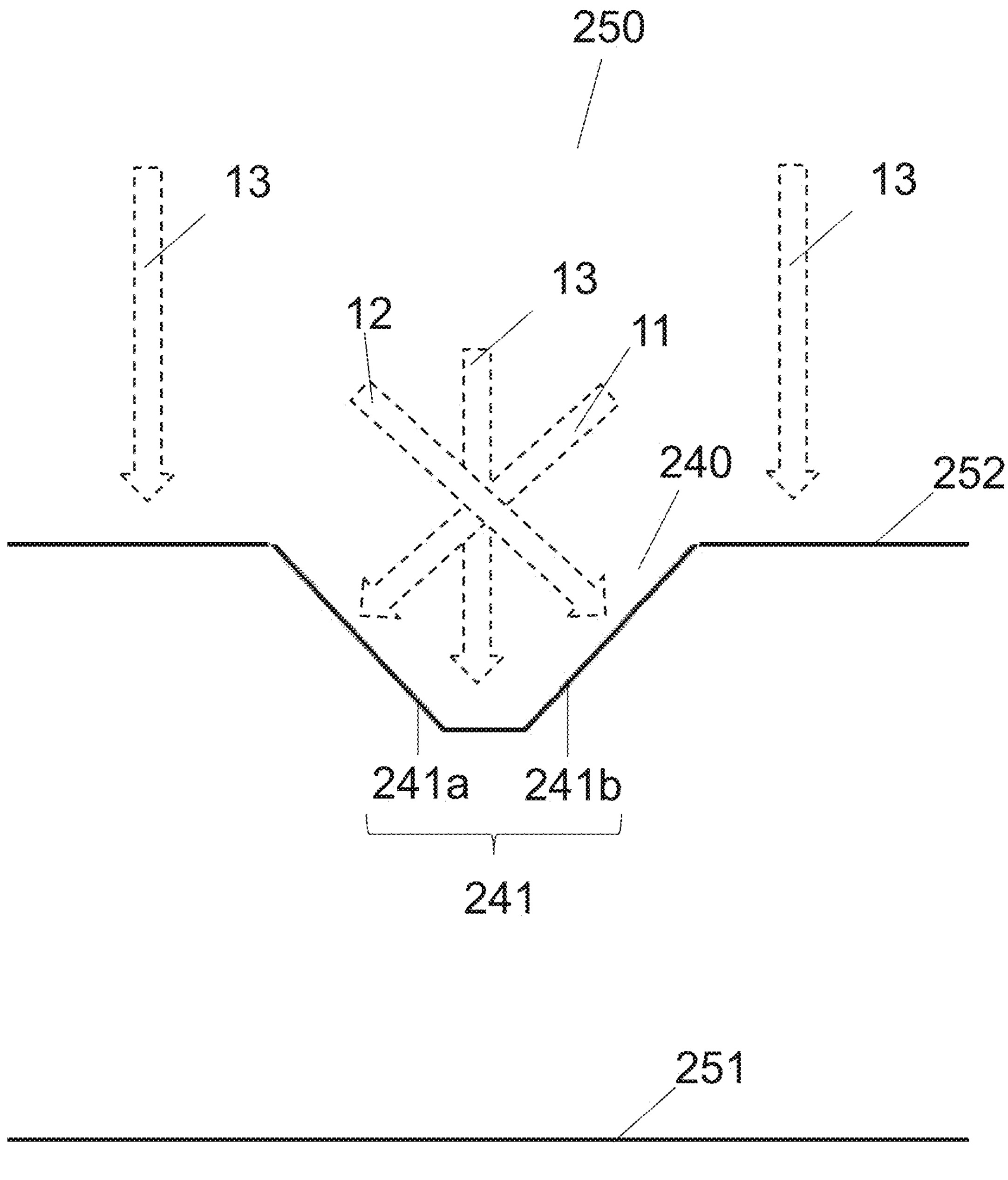
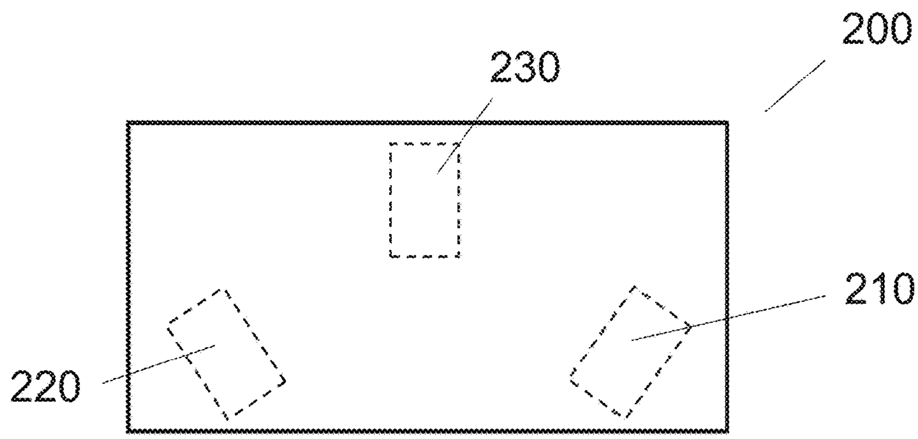


FIG. 2B



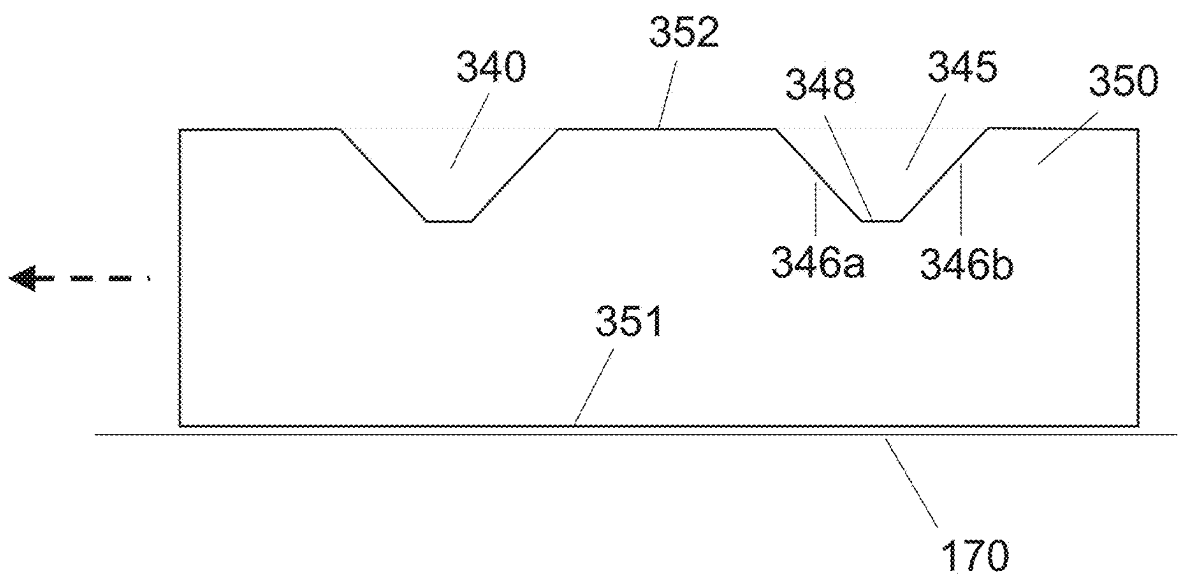


FIG. 3

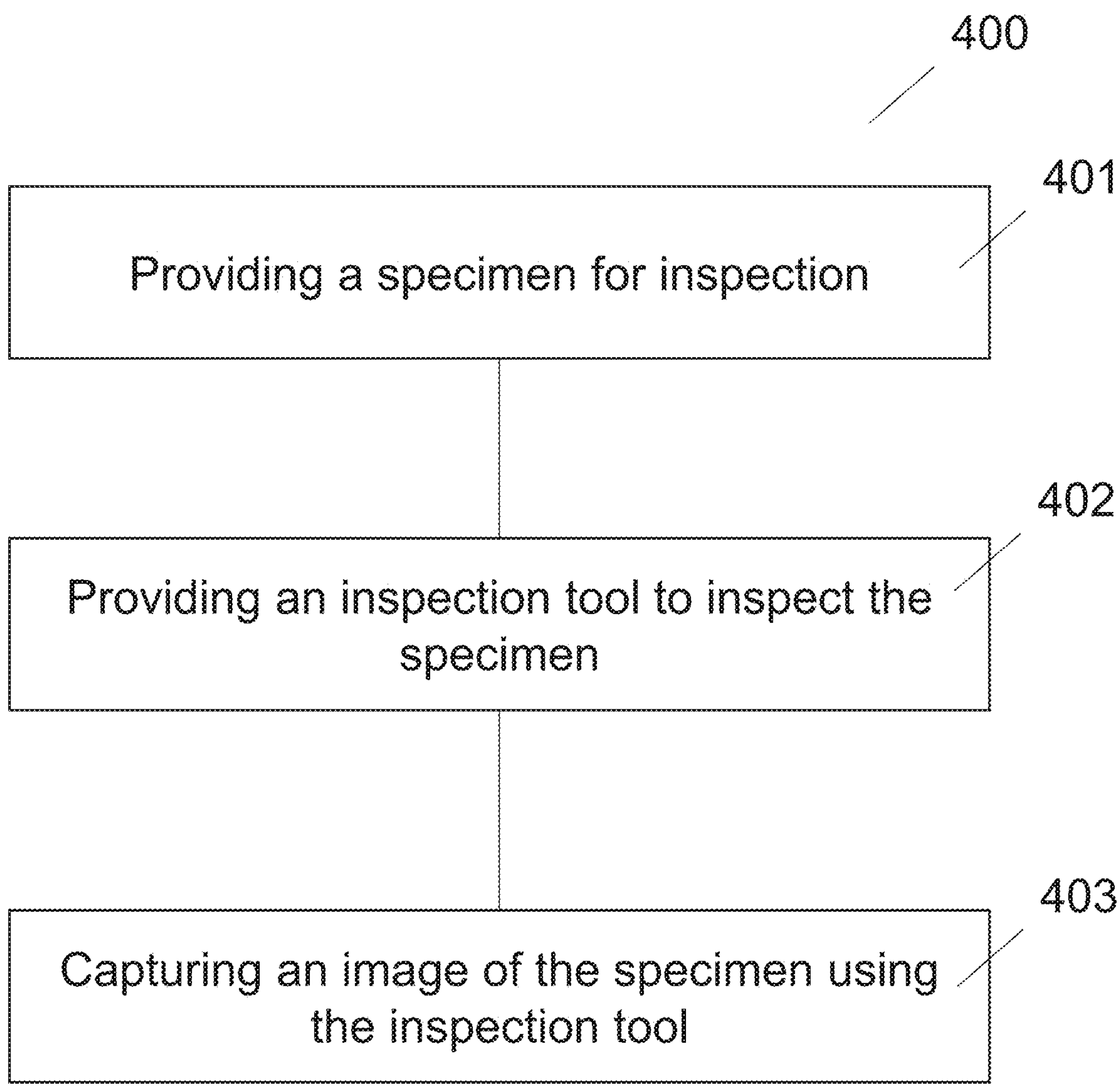

400
401
Providing a specimen for inspection
402
Providing an inspection tool to inspect the specimen
403
Capturing an image of the specimen using the inspection tool

FIG. 4

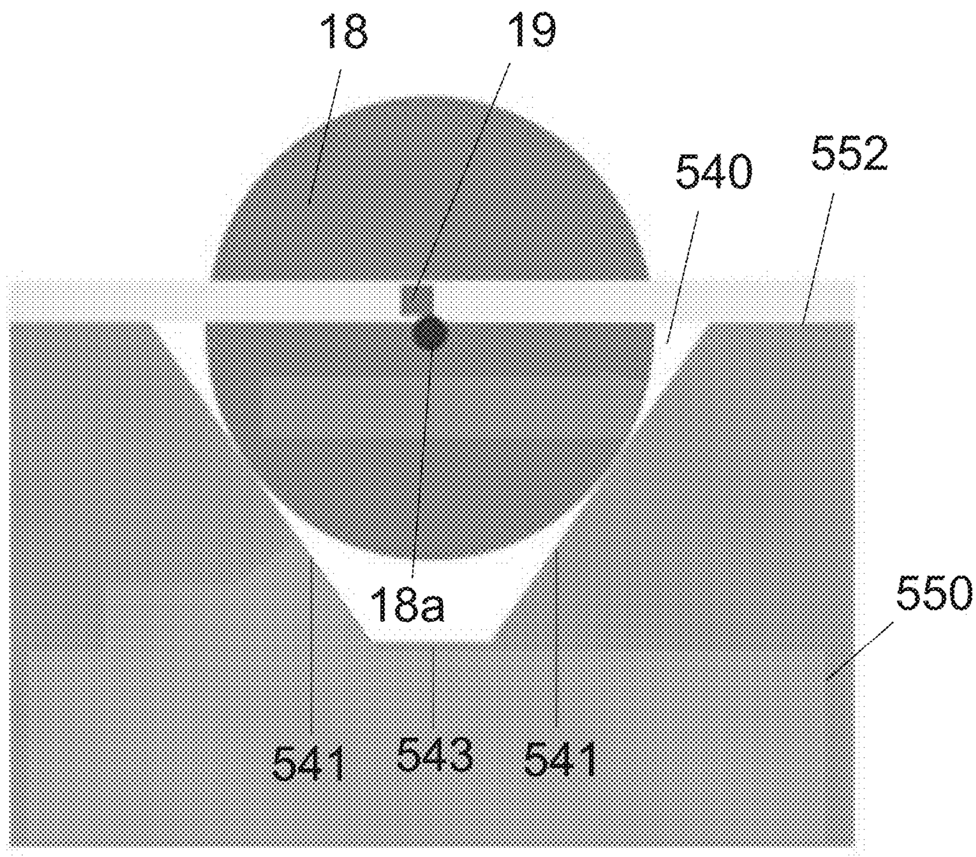


FIG. 5A

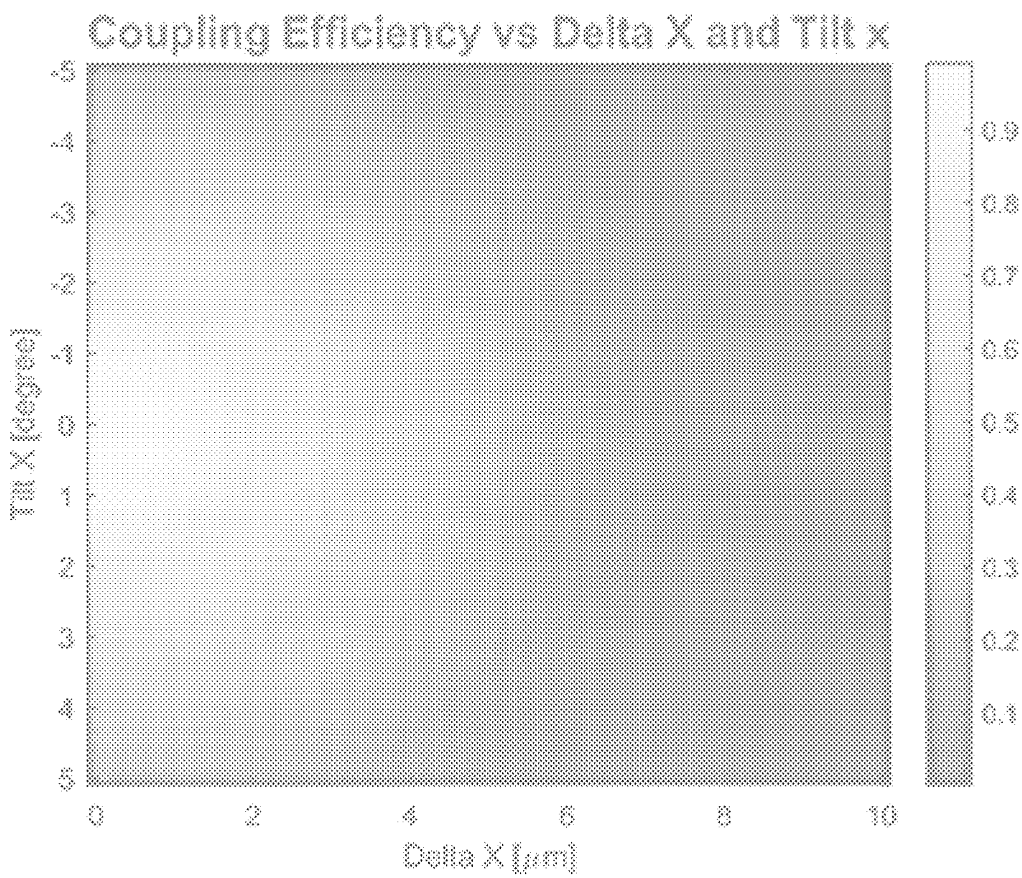

Coupling Efficiency vs Delta X and Tilt x
Tilt X [degree]
-5
-4
-3
-2
-1
0
1
2
3
4
5
0
2
4
6
8
10
Delta X [μm]
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1

FIG. 5B

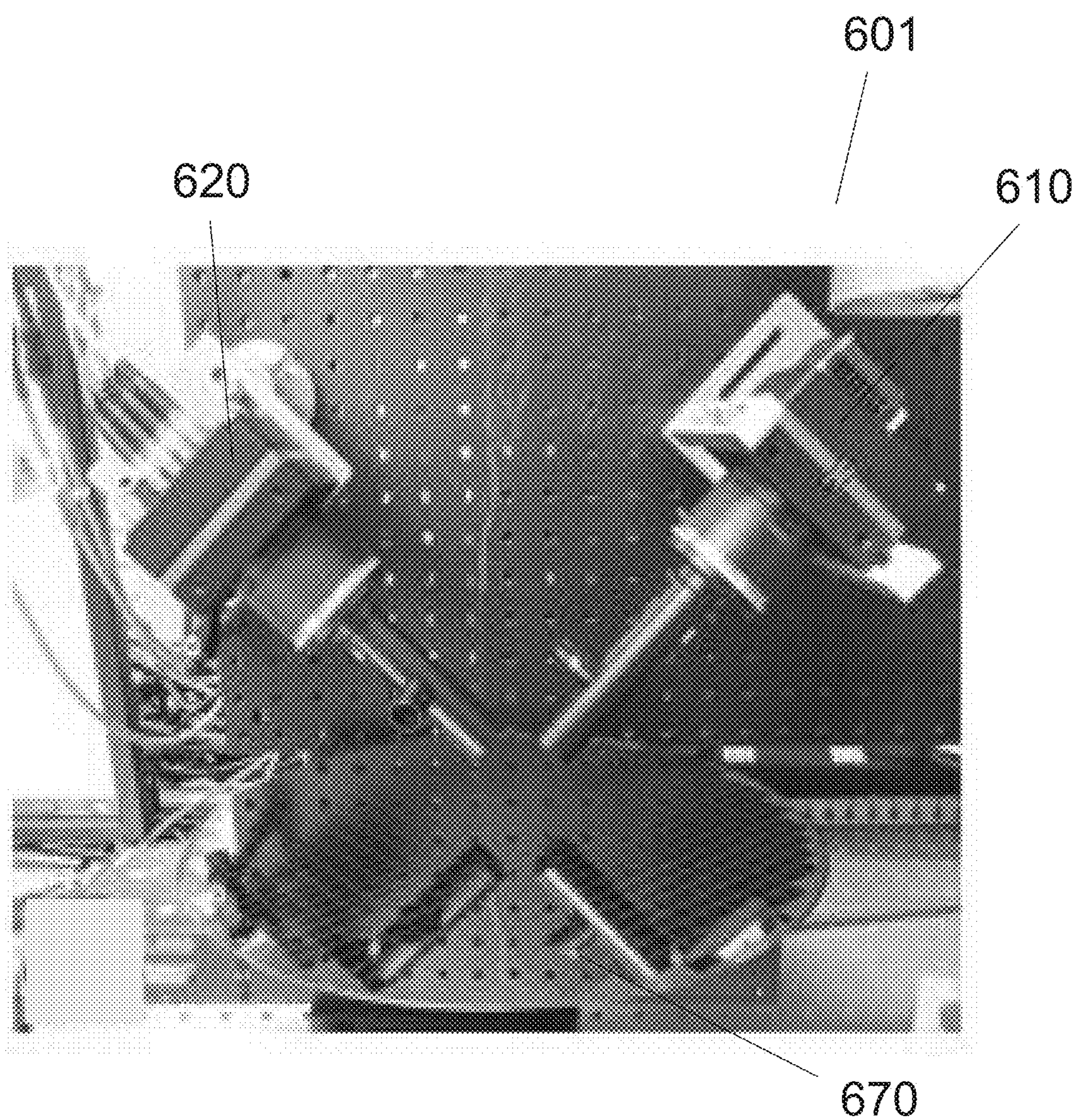


FIG. 6A

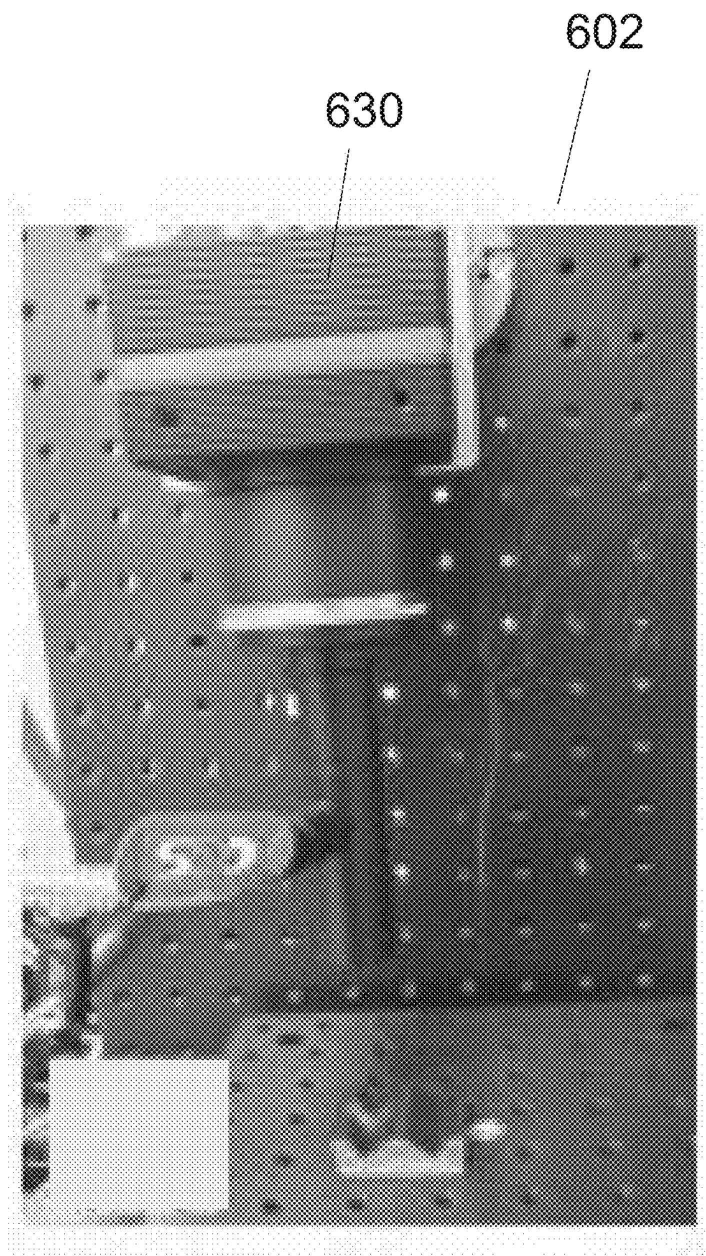


FIG. 6B

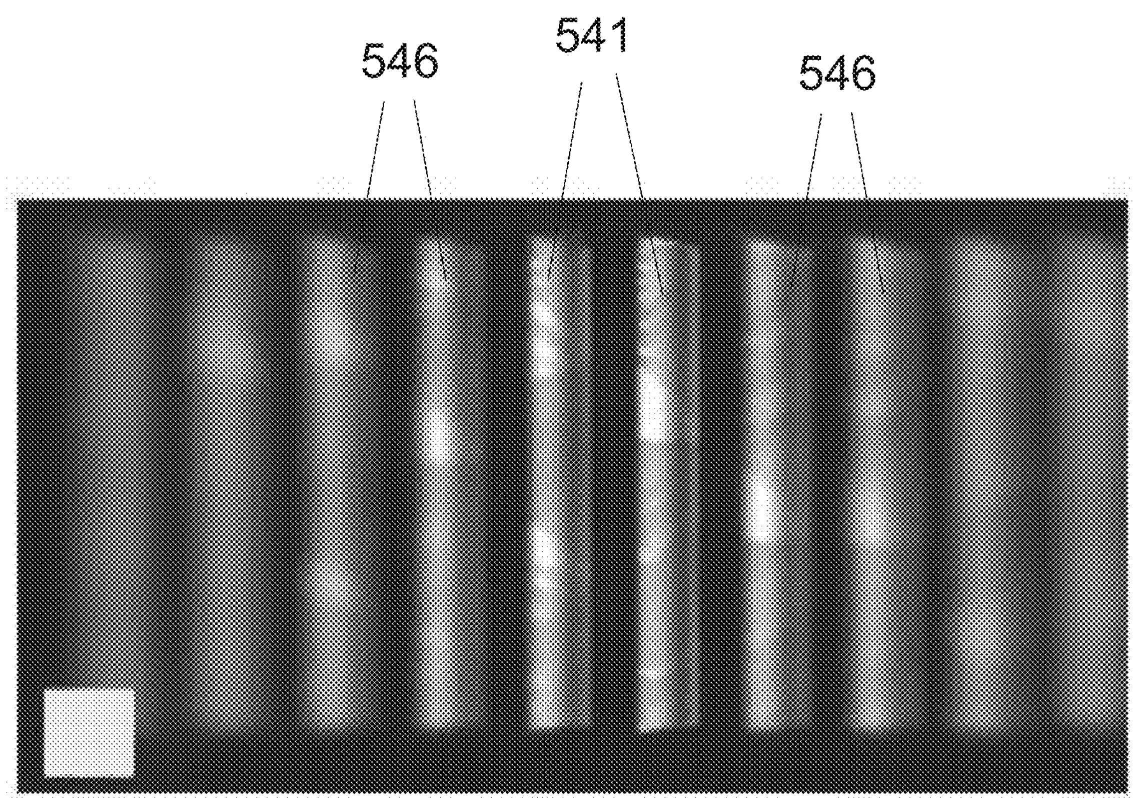


FIG. 7A

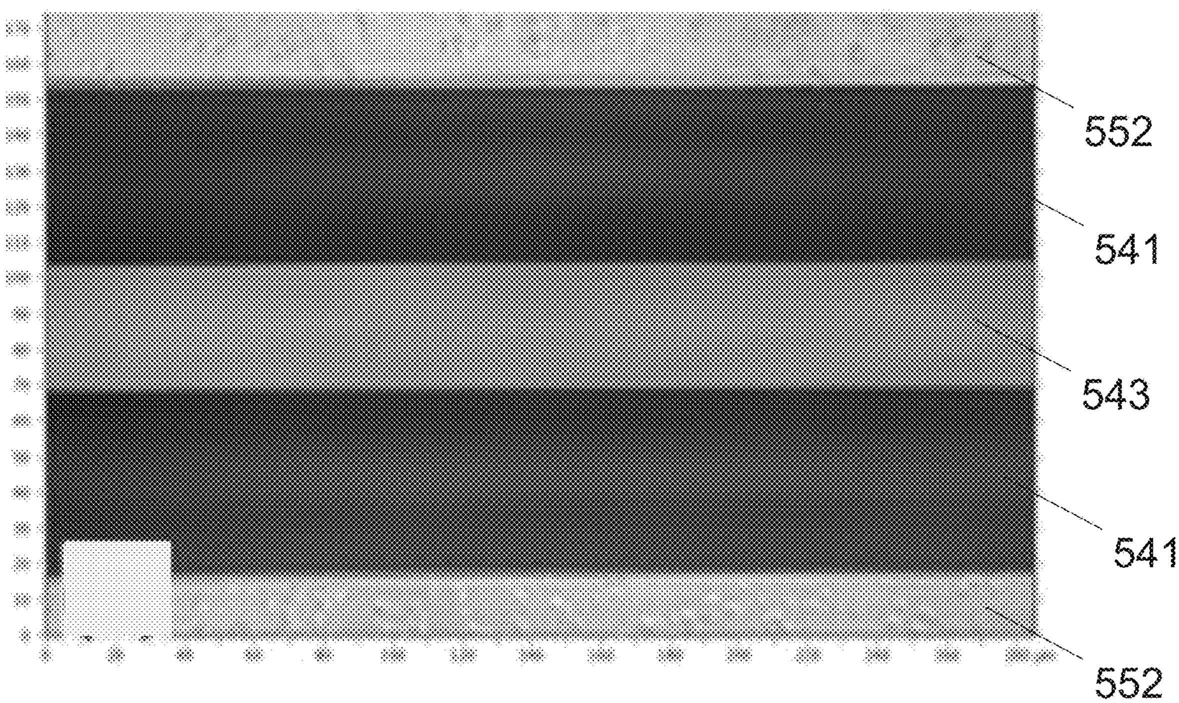


FIG. 7B

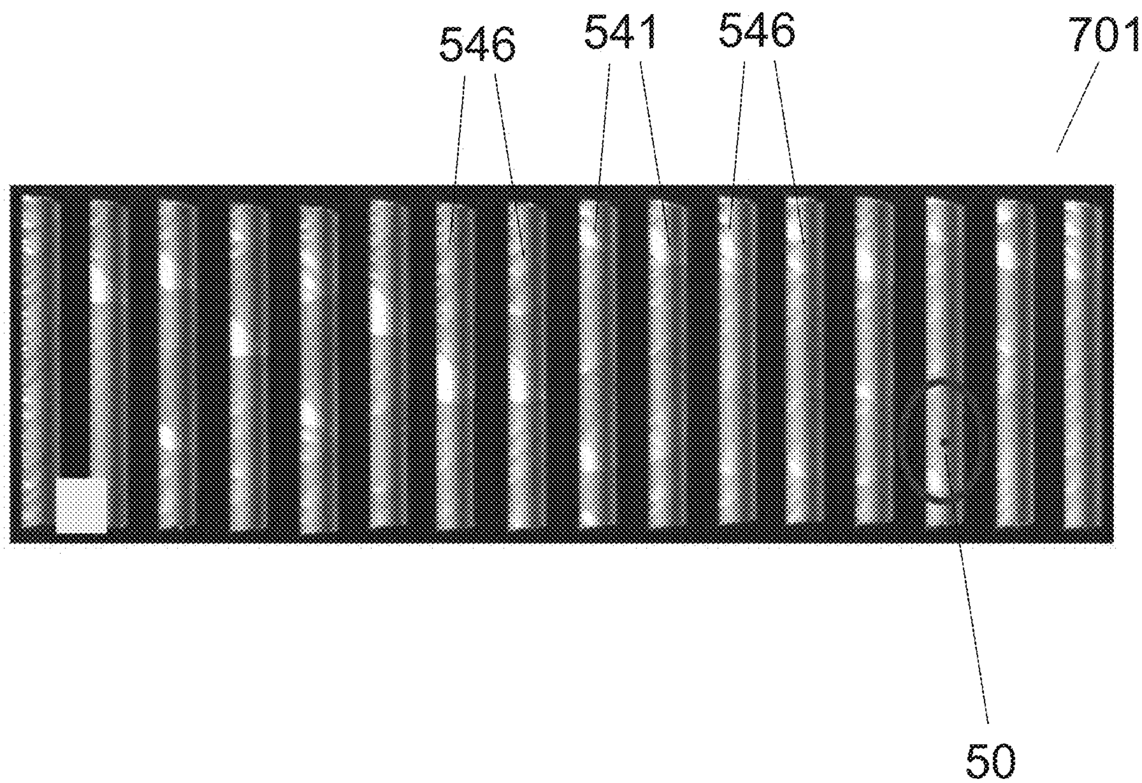


FIG. 7C

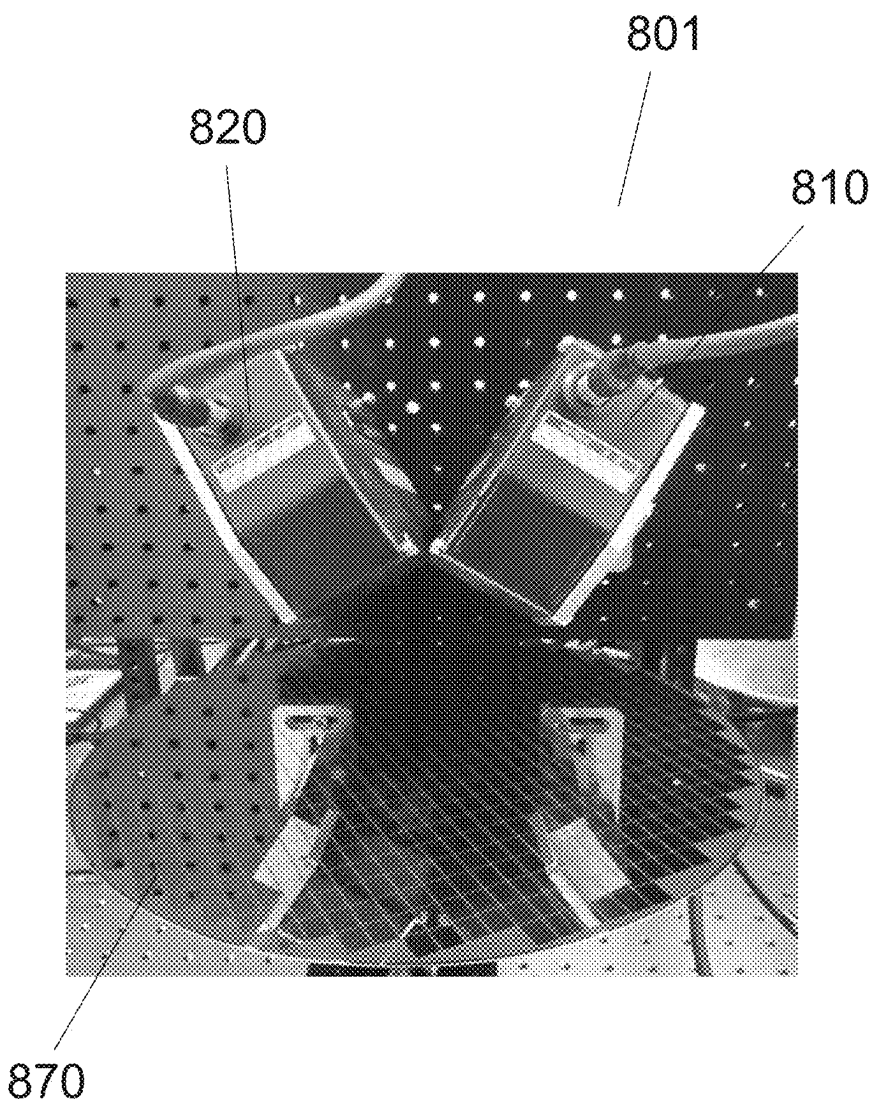


FIG. 8

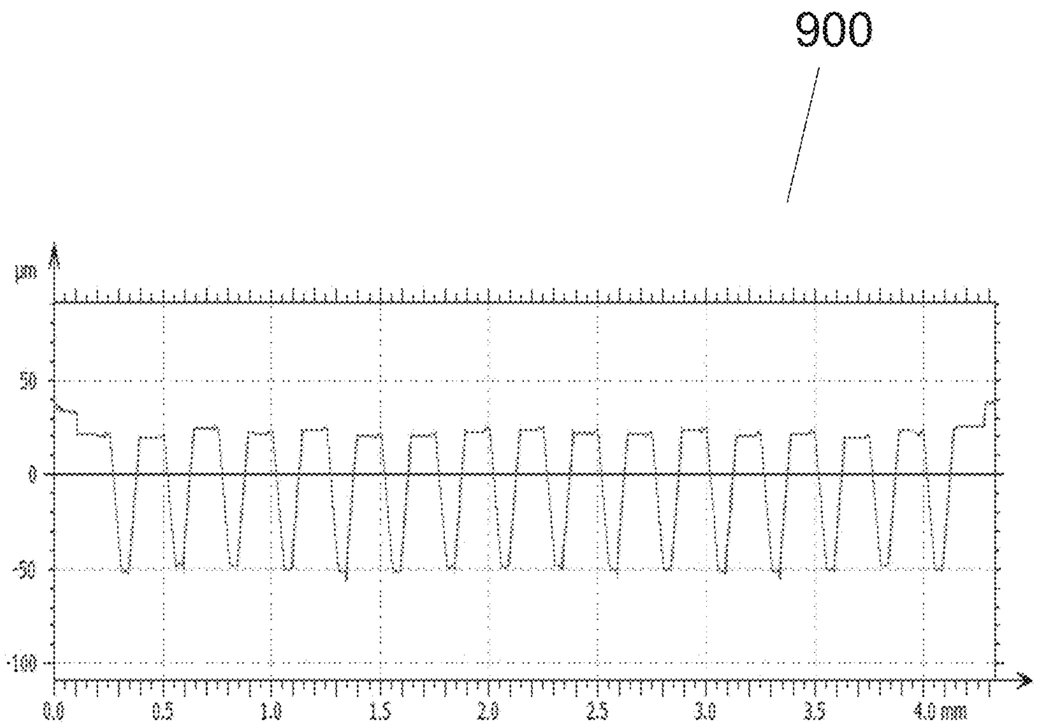


FIG. 9

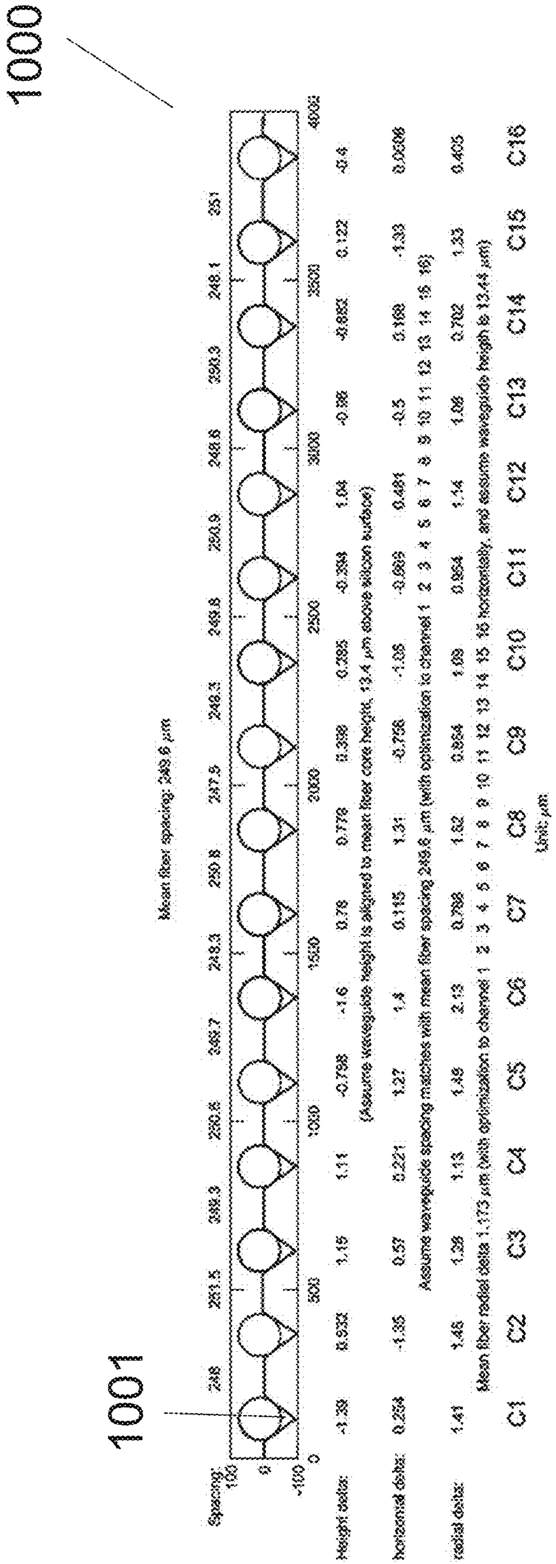


FIG. 10

OPTICAL INSPECTION TOOL AND INSPECTION METHOD FOR INSPECTING MULTIFACETED GROOVE SPECIMENS

TECHNICAL FIELD

Various aspects of the present disclosure generally relate an inspection tool and an inspection method for inspection of a multifaceted channel or groove of a specimen (e.g., workpiece, substrate etc.), and particularly to an inspection tool and an inspection method for inspection of slope-sided grooves or grooved-shaped features or substantially V-shaped grooves in photonics integrated chips (e.g., semiconductor substrates).

BACKGROUND

Butt optical coupling of laser beam(s), which may be generated from silicon photonics chips, coupled to optical fibers placed or fixed on precisely (in other words, high-precision) manufactured slope-sided grooves or V-shaped grooves is a popular approach in the industry (e.g., semiconductor industry). Such V-shaped grooves may be formed or manufactured using processes, such as thin film coating and anisotropic etching, for example, within CMOS semiconductor fabs. A slope of the sidewall(s) of such V-shaped grooves may be defined by the silicon crystal orientation (e.g., of a semiconductor substrate on which the groove(s) may be formed), while an opening/width of the V-shaped grooves may be controlled by a photolithographic mask size. Optical fibers can be guided passively by such V-shaped grooves to be aligned with waveguides/spot size converter (e.g., in photonic chip(s)). However, due to small or tight permissible alignment tolerance (e.g., ~2 μm), care needs to be taken during formation and/or handling of such V-shaped grooves. Foreign materials (e.g., contaminants) in such V-shaped grooves and/or imperfectly etched V-shaped grooves can displace the optical fibers which may, in turn, lead to misalignment between optical fibers and the corresponding waveguides (to which the optical fibers are intended to be aligned with), thereby resulting in low optical coupling efficiency.

Conventional inspection systems and inspection methods are inefficient or slow to conduct and are thus not suitable for high rate or high throughput inspection of V-shaped grooves on semiconductor substrates.

Accordingly, there is a need for an improved inspection tool and inspection method which solve at least the above issue.

BRIEF DESCRIPTION OF THE DRAWINGS

In the drawings, like reference characters generally refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead generally being placed upon illustrating aspects of the disclosure. In the following description, various aspects are described with reference to the following drawings, in which:

FIG. **1** shows a schematic side view of an inspection tool and a specimen for inspection using the inspection tool, according to various aspects of the present disclosure;

FIG. **2**A shows a schematic front view of an inspection tool and a specimen which includes a substantially V-shaped groove, according to various aspects of the present disclosure;

FIG. **2**B shows a close-up schematic front view of surfaces of the specimen of FIG. **2**A which define the substantially V-shaped groove with respect to corresponding orientations or directions of image capture units of the inspection tool of FIG. **2**A, according to various aspects of the present disclosure;

FIG. **3** shows a schematic front view of the inspection tool FIG. **2**A and a specimen which includes a secondary substantially V-shaped groove for inspection by the inspection tool, according to various aspects of the present disclosure.

FIG. **4** depicts an inspection method, according to various aspects of the present disclosure;

FIG. **5**A shows a schematic front view of an optical fiber core, on a substantially V-shaped groove of a specimen, that is misaligned with a waveguide on a silicon PIC;

FIG. **5**B is a graph depicting an optical coupling efficiency's dependence on lateral translation misalignment and angular mismatch;

FIG. **6**A and FIG. **6**B are photographs of prototype inspection tools, according to various aspects of the present disclosure;

FIG. **7**A show an image captured using a first image capture unit and a second image capture unit of the prototype inspection tool of FIG. **6**A, according to various aspects of the present disclosure;

FIG. **7**B show an image captured using a third image capture unit of the prototype inspection tool of FIG. **6**B, according to various aspects of the present disclosure;

FIG. **7**C show a composite image formed by stitching the images captured using the first image capture unit, the second image capture unit, and the third image capture unit of the prototype inspection tool of FIG. **6**A and FIG. **6**B, according to various aspects of the present disclosure;

FIG. **8** is a photograph of another prototype inspection tool, according to various aspects of the present disclosure;

FIG. **9** is a graph, depicting a line profile of substantially V-shaped grooves of a specimen, obtained using at least two laser confocal 3-dimensional profiler of an inspection too; and

FIG. **10** a table depicting a magnitude of any optical fiber core misalignment with respect to a waveguide in a PIC based on the graph of FIG. **9**A.

DETAILED DESCRIPTION

Aspects described below in context of the apparatus are analogously valid for the respective method, and vice versa. Furthermore, it will be understood that the aspects described below may be combined, for example, a part of one aspect may be combined with a part of another aspects.

It should be understood that the terms "on", "over", "top", "bottom", "down", "side", "back", "left", "right", "front", "lateral", "side", "up", "down", etc., when used in the following description are used for convenience and to aid understanding of relative positions or directions, and not intended to limit the orientation of any device, or structure or any part of any device or structure. In addition, the singular terms "a", "an", and "the" include plural references unless context clearly indicates otherwise. Similarly, the word "or" is intended to include "and" unless the context clearly indicates otherwise.

Various aspects generally relate to a multi-vision (e.g., triple-vision) inspection tool or system that may be capable of simultaneously inspecting (i) at least two sidewalls (e.g., angular surfaces) of a V-groove or a substantially V-shaped groove of a specimen and/or (ii) bottom plateau (e.g., floor surface) of the V-groove or substantially V-shaped groove of the specimen and/or (iii) a top surface of the specimen. The inspection tool or system may be used for inspection and/or sorting, at a wafer level of a semiconductor process line, at a singulated die level of the semiconductor process line, and/or at a package level of the semiconductor process line.

In other words, a multi-vision (e.g., double-vision, triple-vision, quadruple-vision etc.) system or inspection tool, which may be capable of simultaneously inspecting or capturing images of a plurality or all surfaces of a V-groove or a substantially V-shaped groove of a specimen may be provided. In particular, three vision systems (e.g., three image capture units) of the multi-vision system or inspection tool may be oriented (e.g., to be perpendicular to each surface of a substantially V-shaped groove of a specimen under inspection) respectively or individually or independently. In addition, each image capture unit of the inspection tool may have very high optical resolution such that the inspection tool may be capable of detecting sub-micrometer foreign materials (e.g., along the substantially V-shaped groove of the specimen).

Accordingly, various aspects may provide an inspection tool (e.g., an optical inspection tool) as well as a specimen for inspection by the inspection tool. The specimen may include a multifaceted channel or groove (in other words, a channel or groove which may be defined by multiple surfaces, e.g., planar surfaces or inner surfaces of the specimen).

The inspection tool according to various aspects may include a plurality of image capture units (e.g., cameras, sensors, 2-dimensional cameras or sensors, 3-dimensional cameras or sensors, 3-dimensional profilers, triangulation sensors, chromatic confocal 3-dimensional profilers, or laser confocal 3-dimensional profilers, etc.) which may be equal to a number of angular surfaces of the channel or groove (e.g., angular or non-parallel with respect to the base surface of the specimen), or may include a number of image capture units equal (e.g., exactly equal) to a total number of facets or surfaces of the channel or groove. Each image capture unit of the inspection tool according to various aspects may be "optimized" to focus or be focused to only one surface (e.g., unique surface) of the multifaceted channel or groove. In another aspect, at least one image capture unit may be "optimized" to focus or be focused to more than one (e.g., two) surfaces (e.g., parallel surfaces).

Various aspects may also provide an inspection method involving the multi-vision (e.g., double-vision, triple-vision, quadruple-vision etc.) system or inspection tool to inspect a specimen. The inspection method according to various aspects may be non-destructive to the specimen under inspection and may not involve or require any contact (e.g., physical or direct contact) between the specimen and the inspection tool. The inspection method may also be performed swiftly. For instance, with the inspection method according to various aspects, all inner surfaces of a specimen defining or forming a V-groove or substantially V-shaped groove of the specimen may be inspected (e.g., have images thereof captured), simultaneously or instantaneously using the inspection tool. In an aspect, the inspection method may be completed within a fraction of second.

Additionally, the inspection method can be implemented at any one or more stage(s) of a process line (e.g., manufacturing line, semiconductor process line, etc.). Accordingly, within a semiconductor process line, the inspection method can be applied or easily conducted at a wafer level, die level, and/or package level of the process.

FIG. **1** shows a schematic side view of an inspection tool **100** and a specimen **150** for inspection using the inspection tool **100**, according to various aspects of the present disclosure.

According to an aspect of the present disclosure, the specimen **150** may be, for example, a workpiece (e.g., metallic workpiece which may be or which may have already been worked on with a tool or a machine, or a metallic part of a ship or vehicle), a substrate (e.g., semiconductor substrate, silicon substrate etc.), a wafer, an electronic circuit board etc., or any other suitable type of specimen for inspection using the inspection tool **100**.

For ease of illustration, various aspects of the disclosure may be described herein with reference to the specimen **150** being a semiconductor substrate. Nevertheless, it is understood that the inspection tool **100** as well as inspection method (described later with reference to FIG. **4**), as disclosed herein, according to various aspects of the present disclosure, may be applicable to or may extended to other types of specimens, for example, a workpiece (e.g., small-sized metallic workpiece, or large-sized workpiece which may be part of a ship or vehicle).

As shown in FIG. **1**, the specimen **150** may include a base surface **151** (e.g., base) and a top surface **152** opposite the base surface **151**. The base surface **151** may be a bottom of the specimen **150** configured to abut or contact or rest on an external surface or platform **170** on which the specimen **150** may also be placed on for inspection by the inspection tool **100**. The top surface **152** of the specimen **150** may be an upper surface of the specimen **150**, opposite the base surface **151**. According to an aspect of the present disclosure, the specimen **150** may further include at least one or a plurality of side surfaces **153***a* and/or **153***b* extending between the base surface **151** and the top surface **152**. Accordingly, the specimen **150** may be shaped as a block or may have a block-like shape, or a non-block-like shape.

According to an aspect of the present disclosure, the specimen **150** may include a first channel **140** for receiving an object therein or therewithin. As shown, the first channel **140** may be positioned in the top surface **152**. The first channel **140** may extend into the specimen **150** from the top surface **152** (e.g., downwards, or towards the base surface **151**). Further, the first channel **140** may run along the top surface **152** of the specimen **150**. Specifically, in an aspect, the first channel **140** may be extending (or running) along the top surface **152** of the specimen **150** in a first direction (e.g., a horizontal/lateral direction, e.g., with respect to the base surface **151** of the specimen **150** or the external surface **170** on which the specimen **150** may be placed), between a first longitudinal end (e.g., first side surface **153***a*) of the specimen **150** and a second longitudinal end opposite the first longitudinal end (e.g., second side surface **153***b* opposite the first side surface **153***a*) of the specimen **150**.

According to an aspect of the present disclosure, the first channel **140** may be a groove. The first channel **140** (e.g., groove) may be a multifaceted groove defined or formed by two or more inner surfaces (e.g., planar or substantially planar inner surfaces) of the specimen **150**.

According to an aspect of the present disclosure, the first channel **140** may be defined or formed by at least two (e.g., two or more than two) angular surfaces **141***a* and **141***b* of the specimen **150**. Specifically, the at least two angular surfaces **141***a* and **141***b* may be angular inner surfaces of the specimen **150** (e.g., opposite outer surfaces, e.g., side surfaces **153***a* and **153***b* of the specimen **150**). The at least two angular surfaces **141***a* and **141***b* may be non-parallel (in other words, may be inclined) with respect to the base surface **151** (e.g., a substantially planar or flat base surface **151**) of the specimen **150**. Specifically, the at least two angular surfaces **141***a* and **141***b* may be declining from the top surface **152**. In other words, the at least two angular surfaces **141***a* and **141***b* may be non-horizontal surfaces and/or may be inclined surfaces with respect to the external surface **170** on which the specimen **150** may be placed. The at least two angular surfaces **141***a* and **141***b* may be disparate and/or different surfaces of the specimen **150** which define the first channel **140** of the specimen **150**. Hence, the at least two angular surfaces **141***a* and **141***b* may be oriented (e.g., with respect to the base surface **151** of the specimen **150** and/or the external surface **170** on which the specimen **150** may be placed) in a different orientation from one another.

In an aspect, the at least two angular surfaces **141***a* and **141***b* of the specimen **150** defining the first channel **140** may be connected to or abutting each other, for instance, at a bottom or a trough of the first channel **140**.

In another aspect, the first channel **140** may further be defined or formed by a floor surface **143** of the specimen **150**. Specifically, the floor surface **143** may be an inner floor surface **143** of the specimen **150** defining the first channel **140**, and/or may be a bottom surface of the first channel **140** (e.g., opposite the outer base surface **151** of the specimen **150**). The floor surface **143** may be positioned or arranged between the at least two angular surfaces **141**. For example, the at least two angular surfaces **141***a* and **141***b* may be adjoined to and may extend (e.g., upwardly, e.g., perpendicularly or non-perpendicularly) from the floor surface **143** of the specimen **150** (e.g., towards or to an opening of the channel at or on the top surface **152** of the specimen **150**). In an aspect, the floor surface **143** may be parallel (e.g., substantially parallel) with the base surface **151** of the specimen **150**. In another aspect, the floor surface **143** may be non-parallel with the base surface **151** of the specimen **150**. Accordingly, according to an aspect of the present disclosure, the first channel **140** may include or may have a shape of a truncated (e.g., half of an) octagon (e.g., regular or irregular octagon), a truncated nonagon (e.g., regular or irregular octagon), a truncated decagon (e.g., regular or irregular octagon) etc.

According to an aspect of the present disclosure, the first channel **140** may be formed by an etching process applied on or to the top surface **152** of the specimen **150**. Hence, when the specimen **150** is a semiconductor substrate or a silicon substrate, an etching process may be applied on or to the top surface **152** of the semiconductor substrate or silicon substrate to form the first channel **140**.

Specifically, according to an aspect of the present disclosure, the first channel **140** may (e.g., through an etching process) be formed as a substantially V-shaped groove or V-like groove (herein collectively or interchangeably referred to as "V-groove") on the specimen **150** (e.g., semiconductor substrate). Such a V-groove (e.g., on a semiconductor substrate or a silicon substrate) may guide and support an optical fiber (i.e. the object) and may serve to align an optical fiber core of the optical fiber (i.e. the object) to a corresponding waveguide (e.g., in a silicon photonic chip (PICs)).

According to an aspect of the present disclosure, a first angular surface **141***a* of the at least two angular surfaces **141***a* and **141***b* of the specimen **150** defining the first channel **140** (e.g., V-groove) may be opposing (e.g., substantially opposing) a second angular surface **141***b* of the at least two angular surfaces **141***a* and **141***b* of the specimen **150** defining the first channel **140**. Further, each of the first angular surface **141***a* and the second angular surface **141***b* of the first channel **140** (e.g., V-groove) may form an angle (e.g., opposing angle) of between substantially 50 degrees to substantially 60 degrees with respect to the base surface **151** of the specimen **150** and/or with respect to the external surface **170** (e.g., when the specimen **150** is placed thereon). Specifically, each of the first angular surface **141***a* and the second angular surface **141***b* of the first channel **140** (e.g., V-groove) may form an angle (e.g., opposing angle) of approximately or substantially 54 to substantially 55 degrees (e.g., 54.74 degrees) with respect to the base surface **151** of the specimen **150** and/or with respect to the external surface **170** (e.g., when the specimen **150** is placed thereon). Thus, for example when the first angular surface **141***a* and the second angular surface **141***b* of the first channel **140** are substantially symmetrical about a central axis or plane of the first channel **140**, the first angular surface **141***a* and the second angular surface **141***b* of the first channel **140** may form an angle (e.g., opposing angle) of between substantially 60 degrees to substantially 80 degrees with respect to one another, or more specifically, between substantially 70 degrees to substantially 72 degrees (e.g., 70.52 degrees) with respect to one another.

According to an aspect of the present disclosure, an opening on the top surface **152** of the specimen **150** for providing access to the first channel **140** (i.e. the opening of the first channel **140** e.g., V-groove) may have a width (e.g., measured laterally) within a range of approximately 100 μm to 200 μm (e.g., 160 μm). In another aspect, the first channel may have a shortest width of not more than or less than 200 μm. Nevertheless, the width of the opening of the first channel **140** is not limited as such. For example, according to various other aspects, the width of the opening of the first channel **140** may be larger than 200 μm.

According to an aspect of the present disclosure, the specimen **150** may further include at least one (e.g., one or more) secondary channel(s) (e.g., further channel(s)) (not shown in FIG. **1**; described in detail later with reference to FIG. **3**). Each secondary channel may be similar or identical to the first channel **140**. That is, each secondary channel may have a similar or an identical dimension, size, shape, form etc., as the first channel. Accordingly, each secondary channel may be positioned in and may extend along the top surface **152** of the specimen **150**, in a same (e.g., identical or similar) direction (i.e. the first direction) as the first channel **140**. Further, the secondary channel may be positioned adjacent or besides or in a side-by-side arrangement with the first channel **140**.

With reference to FIG. **1**, the inspection tool **100** may include (e.g., further include) a support frame **180** configured to or for securing or affixing or supporting the inspection tool **100** to or on the external surface **170**. According to an aspect of the present disclosure, the inspection tool **100** may (e.g., optionally) include a support platform for supporting the specimen **150** thereon. According to an aspect of the present disclosure, the support platform may be included in or may be the external surface **170**.

According to an aspect of the present disclosure, the inspection tool **100** may be an optical inspection tool **100**. The inspection tool **100** (e.g., optical inspection tool **100**) may include at least a first image capture unit **110** and a second image capture unit **120** respectively configured to or may be for inspecting the specimen **150** (e.g., a V-groove of the specimen **150**).

According to an aspect of the present disclosure, the inspection tool **100** (e.g., optical inspection tool **100**) may include (e.g., further include) a third image capture unit **130** configured to or for inspecting the specimen **150** (e.g., V-groove of the specimen **150**).

In an aspect, each or all image capture unit(s) **110** and/or **120** and/or **130** of the inspection tool **100** may be or may include a camera, sensor, 2-dimensional camera or sensor, 3-dimensional camera or sensor, 3-dimensional profiler, triangulation sensor, chromatic confocal 3-dimensional profiler, or laser confocal 3-dimensional profiler, etc.

The first image capture unit **110** may be arranged in a first orientation (e.g., with respect to the base surface **151** of the specimen **150** and/or with respect to the support frame **180** of the inspection tool **100**) and/or may be directable so as to face or be directed towards the first angular surface **141***a* of the first channel **140** or a first secondary angular surface of the secondary channel (e.g., V-groove(s)) (not shown in FIG. **1**) of the specimen **150**. In other words, the first image capture unit **110** when arranged in the first orientation may be directable towards the first angular surface **141***a* of the first channel **140** or the first secondary angular surface of the secondary channel (not shown in FIG. **1**). In particular, according to an aspect of the present disclosure, the first image capture unit **110** may be arranged in the first orientation with respect to the support frame **180** of the inspection tool **100**, such that when the inspection tool **100** is secured to the external surface **170** and the specimen **150** is placed on the external surface **170**, the first image capture unit **110** may be directed towards the first angular surface **141***a* of the first channel **140** or the first secondary angular surface of the secondary channel (not shown in FIG. **1**). Accordingly, the first image capture unit **110** may be configured to capture image(s) of (e.g., of defects and/or contamination on) the first angular surface **141***a* or first secondary angular surface (not shown in FIG. **1**).

The second image capture unit **120** may be arranged in a second orientation (e.g., with respect to the base surface **151** of the specimen **150** and/or with respect to the support frame **180** of the inspection tool **100**) and/or may be directable so as to face or be directed towards the second angular surface **141***b* of the first channel **140** or a second secondary angular surface of the secondary channel (e.g., V-groove(s)) (not shown in FIG. **1**) of the specimen **150**. Hence, the second orientation may differ from the first orientation (e.g., with respect to the base surface **151** of the specimen **150** and/or with respect to the support frame **180** of the inspection tool **100**). In particular, according to an aspect of the present disclosure, the second image capture unit **120** may be arranged in the second orientation with respect to the support frame **180** of the inspection tool **100**, such that when the inspection tool **100** is secured to the external surface **170** and the specimen **150** is placed on the external surface **170**, the second image capture unit **120** may be directed towards the second angular surface **141***b* of the first channel **140** or the second secondary angular surface of the secondary channel. Accordingly, the second image capture unit **120** may be configured to capture image(s) of (e.g., of defects and/or contamination on) the second angular surface **141***b* or second secondary angular surface.

The third image capture unit **130** may be arranged in a third orientation (e.g., with respect to the base surface **151** of the specimen **150** and/or with respect to the support frame **180** of the inspection tool **100**) and/or may be directable so as to face or be directed towards the floor surface **143** of the first channel **140** or a secondary floor surface of the secondary channel (e.g., V-groove(s)) (not shown in FIG. **1**). Hence, the third orientation may differ from the first orientation and the second orientation. In particular, according to an aspect of the present disclosure, the third image capture unit **130** may be arranged in the third orientation with respect to the support frame **180** of the inspection tool **100**, such that when the inspection tool **100** is secured to the external surface **170** and the specimen **150** is placed on the external surface **170**, the third image capture unit **130** may be directed towards the floor surface **143** of the first channel **140** or the secondary floor surface of the secondary channel. Accordingly, the third image capture unit **130** may be configured to capture image(s) of (e.g., of defects and/or contamination on) the floor surface **143** or secondary floor surface.

The inspection tool **100** may (e.g., optionally) include (e.g., further include) a processor **88**. The processor **88** may be coupled to at least the first image capture unit **110** and the second image capture unit **120** and may optionally be further coupled to the third image capture unit **130**. Hence, the processor may be configured to analyse the captured images by the first image capture unit **110** and the second image capture unit **120** and by the third image capture unit **130** (e.g., when the processor is also coupled to the third image capture unit **130**).

Any defects and/or contamination along the first channel **140** or secondary channel (not shown in FIG. **1**) may be identified, for example, by comparing the captured image(s) (e.g., by the first image capture unit **110**, second image capture unit **120**, third image capture unit **130** etc.) against ideal image(s) (e.g., of an ideal angular surface, ideal secondary angular surface, ideal floor surface, ideal secondary floor surface etc., which, for example, may be stored in a memory of the processor **88** or fed to the processor **88** or provided to a human operator) to identify "differences" (e.g., using the processor **88**, or manually using sight of the human operator) between the captured image(s) (e.g., by the first image capture unit **110**, second image capture unit **120**, third image capture unit **130** etc.) and the ideal image(s) as potential defects and/or contamination. For example, defects such as lateral translation misalignment (or lateral misalignment) may be identified by comparing a captured image of a respective surface defining the first channel **140** or secondary channel against an ideal image of the corresponding surface, and any lateral deviation of the captured image from the ideal image may be identified as a lateral translation misalignment. On the other hand, contamination (e.g., contaminants or external particles) may be identified as a dark (or darker) spot or dot (see ref **50** in FIG. **7**C) within a captured image of the respective surface defining the first channel **140** or secondary channel.

According to an aspect of the present disclosure, each of the first image capture unit **110** and/or the second image capture unit **120** and/or the third image capture unit **130** may be "optimized", that is, may be configured to be "focused" on or to (e.g., to only) a surface defining the first channel **140** or the secondary channel that the said image capture unit is directed towards or is facing such that the said image capture unit is capable or is configured to capture a focused or sharp (e.g., substantially sharp) image of (e.g., of only) the said surface.

According to an aspect of the present disclosure, one or more or all of the first image capture unit **110** and/or the second image capture unit **120** and/or the third image capture unit **130** may have a numerical aperture of 0.95 (e.g., substantially 0.95), or a numerical aperture of less than 0.95. For example, each of the first image capture unit **110** and the second image capture unit **120** may have a numerical aperture of less than 0.95, while the third image capture unit **130** may have a numerical aperture of 0.95 (in other words, the numerical aperture of the third image capture unit **130** may be larger/higher in value than the numerical apertures of the first image capture unit **110** and the second image capture unit **120**). As another example, all of the first image capture unit **110**, the second image capture unit **120**, and the third image capture unit **130** may have a same (e.g., identical or similar) numerical aperture (e.g., either 0.95 or of a value less than 0.95).

According to an aspect of the present disclosure, any one or more or all of the first image capture unit **110**, the second image capture unit **120**, and/or the third image capture unit **130** may be configured with a pixel resolution of substantially 500 nm.

Further, any one or more or all of the first image capture unit **110**, the second image capture unit **120**, and/or the third image capture unit **130** may be configured with a field of view of substantially 3.5 mm by 5 mm

Further, any one or more or all of the first image capture unit **110**, the second image capture unit **120**, and/or the third image capture unit **130** may be configured with a depth of field of substantially 30 μm.

According to an aspect of the present disclosure, all image capture units (e.g., the first image capture unit **110**, the second image capture unit **120**, the third image capture unit **130** etc.) of the inspection tool **100** may be identical to each other. That is, according to an aspect of the present disclosure, all image capture units (e.g., the first image capture unit **110**, the second image capture unit **120**, the third image capture unit **130** etc.) of the inspection tool **100** may have a same "image capture" setting and/or configuration and/or may be of a same type of image capture unit (e.g., microscope, digital camera, camera, sensor, 2-dimensional camera or sensor, 3-dimensional camera or sensor, 3-dimensional profiler, triangulation sensor, chromatic confocal 3-dimensional profiler, or laser confocal 3-dimensional profiler etc.).

According to an aspect of the present disclosure, the inspection tool **100** may be part of an inspection system (not shown). In other words, an inspection system (not shown) may include the inspection tool **100** as described herein. The inspection system may further include a first station (e.g., corresponding to a wafer level of a semiconductor process line). The inspection system may further include a second station (e.g., corresponding to a die level of the semiconductor process line). The inspection system may further include a third station (e.g., corresponding to a package level of the semiconductor process line). The second station may be downstream of the first station, and the third station may be downstream of the second station. Each station may include at least one inspection tool **100** (e.g., secured to a corresponding external surface or platform **170** at the station for placing the specimen **150** thereon). Hence, when the inspection system includes three stations, the inspection system may include three inspection tools **100** (i.e. one inspection tool **100** for each station).

In an aspect, the inspection system may include (e.g., further include) the specimen **150**.

In an aspect, the inspection system may include (e.g., further include) a conveyor (not shown) configured to or for moving the specimen **150** between stations (e.g., from the first station to the second station and/or from the second station to the third station), such that at each station, the conveyor positions the specimen **150** (e.g., on a corresponding external surface or platform **170** disposed at that station) in a manner such that each image capture unit of the inspection tool **100** at that station is directed to face a distinct and/or only one inner surface (e.g., angular surface or floor surface) of a channel (e.g., first channel **140** or secondary channel) of the specimen **150**. Further, at each station, the conveyor may also be configured to move the specimen **150** (e.g., with respect to the station), in a manner (e.g., lateral or sideway manner, relative to direction of the conveyor from one station to another station) such that the inspection tool **100** at that station may be capable of capturing image(s) of any secondary channel(s) of the specimen **150** after capturing image(s) of the first channel **140**, for example, without having to move the inspection tool **100** itself.

According to an aspect of the present disclosure, the inspection system (as described) may be positioned or disposed within in an atmospherically-controlled chamber or room, and further, the specimen **150** (or a plurality of specimens **150**) may be conveyed (e.g., by the conveyor) to the inspection system for inspection by the inspection tool **100** of the inspection system within the atmospherically-controlled chamber or room.

FIG. **2**A shows a schematic front view of an inspection tool **200** and a specimen **250** which includes a substantially V-shaped groove **240**, according to various aspects of the present disclosure.

According to an aspect of the present disclosure, the inspection tool **200** of FIG. **2**A may contain any one or more or all the features and/or limitations of the inspection tool **100** of FIG. **1**. In the following, the inspection tool **200** is described with like reference characters generally referring to the same or corresponding parts/features of the inspection tool **100** of FIG. **1**. The description of the parts/features made with respect to inspection tool **200** may also be applicable with respect to inspection tool **100**, and vice versa.

Further, according to an aspect of the present disclosure, the specimen **250** of FIG. **2**A may contain any one or more or all the features and/or limitations of the specimen **150** of FIG. **1**. In the following, the specimen **250** is described with like reference characters generally referring to the same or corresponding parts/features of the specimen **150** of FIG. **1**. The description of the parts/features made with respect to specimen **250** may also be applicable with respect to specimen **150**, and vice versa.

As in the inspection tool **100** of FIG. **1**, the inspection tool **200** of FIG. **2**A may include a first image capture unit **210**, a second image capture unit **220**, and a third image capture unit **230**.

As in the specimen **150** of FIG. **1**, the specimen **250** of FIG. **2**A may include a first channel **240**. As shown, the first channel **240** may be a substantially V-shaped groove defined or formed by a floor surface (i.e. inner floor surface) **243** as well as a first angular surface (i.e. angular inner surface) **241***a* and a second angular surface **241***b* extending from the floor surface **243** towards or to an opening of the first channel **240** at a top surface **252** of the specimen **250**.

The first channel **240** of the specimen **250** may be dimensioned to and/or may be sized to receive or fit an optical fiber (e.g., an object) therein or therewithin and/or at least partially grip or sandwich or hold the optical fiber. Accordingly, each of the first angular surface **241***a* and the second angular surface **241***b* of the first channel **240** of the specimen **250** may form an angle (e.g., opposing angle) of between substantially 50 degrees to substantially 60 degrees with respect to a base surface **251** of the specimen **250**. Specifically, each of the first angular surface **241***a* and the second angular surface **241***b* of the first channel **240** may form an angle (e.g., opposing angle) of approximately or substantially 54 to substantially 55 degrees (e.g., 54.74 degrees) with respect to the base surface **251** of the specimen **250**. Hence, according to an aspect of the present disclosure, for example when the first angular surface **241***a* and the second angular surface **241***b* are substantially symmetrical about a central axis or plane of the first channel **240**, the first angular surface **241***a* and the second angular surface **241***b* of the first channel **240** may form an angle (e.g., opposing angle) of between substantially 60 degrees to substantially 80 degrees with respect to one another, or more specifically, between substantially 70 degrees to substantially 72 degrees with respect to one another (e.g., 70.52 degrees) with respect to one another, for receiving (e.g., at least partially receiving and/or holding) the optical fiber therebetween. According to an aspect of the present disclosure, the opening on the top surface **252** of the specimen **250** for providing access to the first channel **240** may have a width (e.g., measured laterally) or a shortest width equal (e.g., substantially equal) to or larger than a largest width of the optical fiber.

FIG. **2**B shows a close-up schematic front view of surfaces of the specimen **250** of FIG. **2**A which define the substantially V-shaped groove **240** with respect to corresponding orientations or directions of the image capture units **210**, **220** and **230** of the inspection tool **250** of FIG. **2**A, according to various aspects of the present disclosure.

With reference to FIG. **2**A and FIG. **2**B, the first image capture unit **210** may be configured in a first orientation or direction (as shown by arrow **11**) to face or be directed to the first angular surface **241***a* of the first channel **240**. In particular, an optical axis of the first image capture unit **210** (e.g., optical axis of a lens, camera, sensor etc. of the first image capture unit **210**) may be perpendicular (e.g., substantially perpendicular) to the first angular surface **241***a*. The first image capture unit **210** may be configured to be focused to (e.g., to only) the first angular surface **241***a* so as to capture an image (e.g., sharp image) of (e.g., only of) the first angular surface **241***a*.

The second image capture unit **220** may be configured in a second orientation or direction (as shown by arrow **12**), which may be different from the first orientation, to face or be directed to the second angular surface **241***b* of the first channel **240**. In particular, an optical axis of the second image capture unit **220** (e.g., optical axis of a lens, camera, sensor etc. of the second image capture unit **220**) may be perpendicular (e.g., substantially perpendicular) to the second angular surface **241***b*. The second image capture unit **220** may be configured to be focused to (e.g., to only) the second angular surface **241***b* so as to capture an image (e.g., sharp image) of (e.g., only of) the second angular surface **241***b*.

The third image capture unit **230** may be configured in a third orientation or direction (as shown by arrow **13**), which may be different from the first orientation and the second orientation, to face or be directed to the floor surface **243** of the first channel **240**. In particular, an optical axis of the third image capture unit **230** (e.g., optical axis of a lens, camera, sensor etc. of the third image capture unit **230**) may be perpendicular (e.g., substantially perpendicular) to the floor surface **243**. In an aspect, the third image capture unit **230** may be configured to be focused to (e.g., to only) the floor surface **243** so as to capture an image (e.g., sharp image) of (e.g., only of) the floor surface **243**. In another aspect, the floor surface **243** may be parallel (e.g., substantially parallel) with the top surface **252** of the specimen **250**, and the third image capture unit **230** may be configured to be focused (e.g., substantially focused) to both the floor surface **243** and the top surface **252** of the specimen **250** (e.g., simultaneously, or alternatively, in sequence) so as to be capable of capturing an image (e.g., a substantially sharp or focused image) of both the floor surface **243** and the top surface **252** of the specimen **250**.

FIG. **3** shows a schematic front view of the inspection tool **200** of FIG. **2**A and a specimen **350** which includes a secondary channel **345** for inspection by the inspection tool **200**, according to various aspects of the present disclosure.

As shown, the specimen **350** may include a first channel **340**. The first channel **340** of the specimen **350** may be similar or identical to the first channel **140** of the specimen **150** of FIG. **1** or the first channel **240** of the specimen **250** of FIG. **2**A.

As shown, according to an aspect of the present disclosure, the specimen **350** may further include at least one (e.g., one or more) secondary channel(s) **345** (e.g., further channel (s)). Each secondary channel **345** may be similar or identical to the first channel **340**. That is, each secondary channel **345** may have a similar or an identical dimension, size, shape, form etc., as the first channel **340**.

Accordingly, the secondary channel **345** may be positioned in and may extend along a top surface **352** of the specimen **350**, in a same (e.g., identical or similar) direction (i.e. the first direction) as the first channel **340**. Further, the secondary channel **345** may be positioned adjacent or besides the first channel **340**. In other words, the first channel **340** as well as the secondary channel **345** (e.g., at least one or more or all secondary channel(s) **345**) may be in a side-by-side arrangement with respect to one another, on or along the top surface **352** of the specimen **350**. The secondary channel **345** may be defined or formed by at least two secondary (e.g., at least two other) angular surfaces (i.e. angular inner surfaces) **346***a* and **346***b* of the specimen **350**. As shown, the at least two secondary angular surfaces **346***a* and **346***b* defining the secondary channel **345** may be non-parallel (i.e. inclined) with a base surface **351** of the specimen **350**. Further, the at least two secondary angular surfaces **346***a* and **346***b* may be oriented with respect to the base surface **351** of the specimen **350** in a different orientation from one another.

In an aspect, the secondary channel **345** may further be defined or formed by a secondary floor surface **348** of the specimen **350**. The secondary floor surface **348** may be positioned or arranged between the at least two secondary angular surfaces **346***a* and **346***b* defining the secondary channel **345**. For example, the at least two secondary angular surfaces **346***a* and **346***b* may be adjoined to and may extend (e.g., upwardly, e.g., perpendicularly or non-perpendicularly) from the secondary floor surface **348** of the specimen **350** defining the secondary channel **345** (e.g., towards or to an opening of the channel at or on the top surface **352**). In an aspect, the secondary floor surface **348** may be parallel (e.g., substantially parallel) with the base surface **351** (e.g., substantially planar or flat base surface **351**) of the specimen **350**. In another aspect, the secondary floor surface **348** may be non-parallel with the base surface **351** of the specimen **350**. Accordingly, according to an aspect of the present disclosure, the secondary channel **345** may include or may have a shape of a truncated octagon (e.g., regular or irregular octagon), a truncated nonagon (e.g., regular or irregular octagon), a truncated decagon (e.g., regular or irregular octagon) etc.

In another aspect (not shown), the at least two secondary angular surfaces **346***a* and **346***b* defining the secondary channel **345** may be connected to or abutting each other, for instance, at a bottom or a trough of the secondary channel **345**.

According to an aspect of the present disclosure, the secondary channel **345** may be formed by an etching process applied on or to the top surface **352** of the specimen **350** (e.g., a semiconductor substrate or a silicon substrate).

Specifically, according to an aspect of the present disclosure, the secondary channel **345** may (e.g., through an etching process) be formed as a substantially V-shaped groove or V-like groove (i.e. "V-groove") on the specimen **350** (e.g., semiconductor substrate or silicon substrate).

Accordingly, a first secondary angular surface **346***a* of the specimen **350** defining the secondary channel **345** (e.g., V-groove) may be opposing (e.g., substantially opposing) a second secondary angular surface **346***b* of the specimen **350** defining the secondary channel **345**. Further, each of the first secondary angular surface **346***a* and the second secondary angular surface **346***b* defining the secondary channel **345** (e.g., V-groove) may form an angle (e.g., opposing angle) of between substantially 50 degrees to substantially 60 degrees with respect to the base surface **351** of the specimen **350** and/or with respect to the external surface **170** (e.g., when the specimen **350** is placed thereon). Specifically, each of the first secondary angular surface **346***a* and the second secondary angular surface **346***b* defining the secondary channel **345** (e.g., V-groove) may form an angle (e.g., opposing angle) of approximately or substantially 54 to substantially 55 degrees (e.g., 54.74 degrees) with respect to the base surface **351** of the specimen **350** and/or with respect to the external surface **170** (e.g., when the specimen **350** is placed thereon). Thus, for example when the first secondary angular surface **346***a* and the second secondary angular surface **346***b* are substantially symmetrical about a central axis or plane of the secondary channel **345**, the first secondary angular surface **346***a* and the second secondary angular surface **346***b* defining the secondary channel **345** may form an angle (e.g., opposing angle) of between substantially 60 degrees to substantially 80 degrees with respect to one another, or more specifically, between substantially 70 degrees to substantially 72 degrees (e.g., 70.52 degrees) with respect to one another.

To inspect the first channel **340** (e.g., a longitudinal segment of or the entire first channel **340**) using the inspection tool **200**, the specimen **350** may be moved in the first direction (e.g., in a same direction in which the first channel **340** is extending along), with respect the inspection tool **200**, or vice versa (in other words, the inspection tool **200** may be moved relative to the specimen **350**), while the image capture units **210**, **220** and **230** of the inspection tool **200** are facing or directed towards the first channel **340**.

To inspect the secondary channel **345**, for example, after inspecting the first channel **340**, the specimen **350** may be moved (e.g., in a second direction substantially perpendicular to the first direction), relative to the inspection tool **200**, or vice versa (in other words, the inspection tool **200** may be moved relative to the specimen **350**), such that (i) the first image capture unit **210** of the inspection tool **200** may face or be directed towards the first secondary angular surface **346***a* of the secondary channel **345**, (ii) the second image capture unit **220** of the inspection tool **200** may face or be directed towards the second secondary angular surface **346***b* of the secondary channel **345**, and/or (iii) the third image capture unit **230** of the inspection tool **200** may face or be directed towards the secondary floor surface **348** of the secondary channel **345**.

FIG. **4** depicts an inspection method **400**, according to various aspects of the present disclosure.

With reference to FIG. **1**, FIG. **2**A, FIG. **2**B and FIG. **3**, the inspection method **400**, which may include or involve the inspection tool **100** or **200** and/or the specimen **150**, **250** or **350**, may be described.

According to an aspect of the present disclosure, the method **400** may include a step **401** of providing the specimen **150**, **250** or **350** (e.g., semiconductor substrate) for inspection. The specimen **150**, **250** or **350** may include the base surface **151** or **251**, the top surface **152** or **252** opposite the base surface **151** or **251**, and the first channel **140**, **240** or **340** positioned in the top surface **152** or **252** in the first direction for receiving an object therein. The first channel **140**, **240** or **340** may be formed by at least two angular surfaces **141***a* and **141***b* or **241***a* and **241***b* of the specimen **150**, **250** or **350**. The at least two angular surfaces **141***a* and **141***b* or **241***a* and **241***b* may be inclined with the base surface **151** or **251** of the specimen **150**, **250** or **350**. Further, at least two angular surfaces **141***a* and **141***b* or **241***a* and **241***b* may be oriented with respect to the base surface **151** or **251** in a different orientation from one another.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) a step **402** providing the inspection tool **100** or **200** to inspect the specimen **150**, **250** or **350**. The inspection tool **100** or **200** may include (i) the first image capture unit **110** or **210** oriented or configured to face a first angular surface **141***a* or **241***a* of the at least two angular surfaces **141***a* and **141***b* or **241***a* and **241***b* and (ii) the second image capture unit **120** or **220** oriented or configured to face a second angular surface **141***b* or **241***b* of the at least two angular surfaces **141***a* and **141***b* or **241**. The inspection tool **100** or **200** may further include the support platform (e.g., which may be included in or may be the external surface **170**).

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) placing the specimen **150**, **250** or **350** on the support platform (e.g., which may be included in or may be the external surface **170**).

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) a step **403** capturing an image (e.g., a first image) of the first angular surface **141***a* or **241***a* using the first image capture unit **110** or **210** and capturing an image (e.g., a second image) of the second angular surface **141***b* or **241***b* using the second image capture unit **120** or **220** (e.g., after placing the specimen **150**, **250** or **350** on the support platform), for inspection of the specimen **150**, **250** or **350**.

According to an aspect of the present disclosure, in the method **400**, the optical axis of the first image capture unit **110** or **210** may be substantially perpendicular to the first angular surface **141***a* or **241***a*. Further, the optical axis of the second image capture unit **120** or **220** may be substantially perpendicular to the second angular surface **141***b* or **241***b*.

According to an aspect of the present disclosure, in the method **400**, the first channel **140**, **240** or **340** may further include the floor surface **143** or **243** **143** or **243** that may be substantially parallel with the base surface **151** or **251** of the specimen **150**, **250** or **350**. The floor surface **143** or **243** **143** or **243** may be adjoined to and may be positioned between the first angular surface **141***a* or **241***a* and the second angular surface **141***b* or **241***b*.

According to an aspect of the present disclosure, in the method **400**, the inspection tool **100** or **200** may further include the third image capture unit **130** or **230** which may be oriented or configured to face the floor surface **143** or **243** **143** or **243**. According to an aspect of the present disclosure, the method **400** may include (e.g., further include) capturing an image (e.g., a third image) of the floor surface **143** or **243** **143** or **243** using the third image capture unit **130** or **230**.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) stitching the captured first image, second image, and third image to form a composite image of the first channel **140**, **240** or **340**. For example, the first image may be cropped in a manner such that only an image of the first angular surface **141***a* or **241***a* remains (or is shown) (herein referred to as "first cropped image"). Further, the second image may be cropped in a manner such that only an image of the second angular surface **141***b* or **241***b* remains (herein referred to as "second cropped image"). Yet further, the third image may be cropped in a manner such that only an image of the floor surface **143** or **243 143** or **243** remains (herein referred to as "third cropped image"). The method **400** may include stitching the first cropped image, the second cropped image, and the third cropped image (e.g., with the third cropped image between the first and the second cropped images) form the composite image of the first channel **140**, **240** or **340**.

According to another aspect of the present disclosure, the method **400** may include (e.g., further include) stitching the captured first image and second image to form a composite image of the first angular surface **141***a* or **241***a* and second angular surface **141***b* or **241***b* of the first channel **140**, **240** or **340**.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) analysing the captured first image, second image and third image (e.g., using the processor **88**) for identifying defects and/or contamination. In particular, the method **400** may include (e.g., further include) analysing the composite image of the first channel **140**, **240** or **340** or the composite image of the first angular surface **141***a* or **241***a* and second angular surface **141***b* or **241***b* of the first channel **140**, **240** or **340**.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) operating at least the first image capture unit **110** or **210** and the second image capture unit **120** or **220** simultaneously. In particular, the first image capture unit **110** or **210** and the second image capture unit **120** or **220** of the inspection tool **100** or **200** may be configured to operate simultaneously to capture the first image of the first angular surface **141***a* or **241***a* and the second image of the second angular surface **141***b* or **241***b* at a same time.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) operating at least the first image capture unit **110** or **210**, the second image capture unit **120** or **220**, and the third image capture unit **130** or **230** simultaneously. In particular, the first image capture unit **110** or **210**, the second image capture unit **120** or **220**, and the third image capture unit **130** or **230** of the inspection tool **100** or **200** may be configured to operate simultaneously to capture the first image of the first angular surface **141***a* or **241***a*, the second image of the second angular surface **141***b* or **241***b*, and the third image of the floor surface **143** or **243 143** or **243** at a same time.

According to an aspect of the present disclosure, any one or more of the first image capture unit **110** or **210** and/or the second image capture unit **120** or **220** and/or the third image capture unit **130** or **230** may be configured to be operated (e.g., simultaneously) at regular, or irregular, time intervals in order to or for capturing a sequence or particular sequence of events (e.g., temporal-related events, for example, movement of a particle or contaminant across a width of a V-groove).

Accordingly, operating a plurality or all image capture units directed to a channel (e.g., first channel **140**, **240** or **340** or secondary channel **345**) of the specimen **150**, **250** or **350** simultaneously, at time intervals (e.g., regular time intervals), a temporal (e.g., time-related) account (e.g., depiction or understanding) of movement, e.g., of a particle, contamination particle, or a fluid, across the said channel (e.g., first channel **140**, **240** or **340** or secondary channel **345**) of the specimen **150**, **250** or **350** may be established.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) moving the specimen **150**, **250** or **350**, in the first direction (e.g., in a same direction in which the channel is extending along), with respect to at least the first image capture unit **110** or **210** and the second image capture unit **120** or **220** (and, optionally, the third image capture unit **130** or **230**) of the inspection tool **100** or **200**.

According to an aspect of the present disclosure, within the method **400**, the specimen **150**, **250** or **350** may further include the (e.g., at least one) secondary channel **345** for receiving the object (e.g., optical fiber). Accordingly, the secondary channel **345** may be positioned in and/or may be extending along the top surface **152** or **252** (e.g., top surface **152** or **252**) of the specimen **150**, **250** or **350** in the first direction, and the secondary channel **345** may be adjacent to the first channel **140**, **240** or **340**. Further, the secondary channel **345** may be defined by at least two secondary angular surfaces **346** of the specimen **150**, **250** or **350**, the at least two secondary angular surfaces **346** being non-parallel with the base surface **151** or **251** of the specimen **150**, **250** or **350** and may be oriented with respect to the base surface **151** or **251** differently from one another.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) moving the specimen **150**, **250** or **350**, in a second direction substantially perpendicular to the first direction, such that the first image capture unit **110** or **210** of the inspection tool **100** or **200** may face or be directed towards the first secondary angular surface **346***a* of the secondary channel **345** and the second image capture unit **120** or **220** of the inspection tool **100** or **200** may face or be directed towards the second secondary angular surface **346***b* of the secondary channel **345**. The method **400** may further include capturing an image of the first secondary angular surface **346***a* using the first image capture unit **110** or **210** and capturing an image of the second secondary angular surface **346***b* using the second image capture unit **120** or **220**, for example (e.g., after moving the specimen **150**, **250** or **350** in the second direction as described). The method **400** may further include capturing an image of the secondary floor surface **348** using the third image capture unit **130** or **230**.

According to an aspect of the present disclosure, within the method **400**, the specimen **150**, **250** or **350** may be a semiconductor substrate (e.g., silicon substrate).

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) placing the object in the first channel **140**, **240** or **340** after capturing the first image and the second image. According to an aspect of the present disclosure, when the inspection tool **100** or **200** includes the third image capture unit **130** or **230**, the method **400** may include (e.g., further include) placing the object in the first channel **140**, **240** or **340** after capturing the first image, the second image, and the third image.

According to an aspect of the present disclosure, within the method **400**, each of the first channel **140**, **240** or **340** and/or the secondary channel **345** may be formed using an etching process on the semiconductor substrate.

According to an aspect of the present disclosure, within the method **400**, each of the first angular surface **141***a* or **241***a* and the second angular surface **141***b* or **241***b* of the first channel **140**, **240** or **340** may form an opposing angle with the base surface **151** or **251** of the specimen **150**, **250** or **350** of between substantially 50 degrees to substantially 60 degrees. Further, each of the first secondary angular surface **346***a* and the second secondary angular surface **346***b* of the secondary channel **345** may form an opposing angle with the base surface **151** or **251** of the specimen **150**, **250** or **350** of between substantially 50 degrees to substantially 60 degrees.

According to an aspect of the present disclosure, within the method **400**, the first angular surface **141***a* or **241***a* and the second angular surface **141***b* or **241***b* of the first channel **140**, **240** or **340** may form an angle with respect to each other of between substantially 60 degrees to substantially 80 degree. Further, the first secondary angular surface **346***a* and the second secondary angular surface **346***b* of the secondary channel **345** may form an angle with respect to each other of between substantially 60 degrees to substantially 80 degree.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) configuring each image capture unit (e.g., first image capture unit **110** or **210**, second image capture unit **120** or **220**, third image capture unit **130** or **230** etc.) of the inspection tool **100** or **200** to have a numerical aperture of less than 0.95.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) configuring any one or all image capture unit(s) (e.g., first image capture unit **110** or **210**, second image capture unit **120** or **220**, third image capture unit **130** or **230** etc.) of the inspection tool **100** or **200** with a pixel resolution of substantially 500 nm.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) configuring any one or all image capture unit(s) (e.g., first image capture unit **110** or **210**, second image capture unit **120** or **220**, third image capture unit **130** or **230** etc.) of the inspection tool **100** or **200** with a field of view of substantially 3.5 mm by 5 mm.

According to an aspect of the present disclosure, the method **400** may include (e.g., further include) configuring any one or all image capture unit(s) (e.g., first image capture unit **110** or **210**, second image capture unit **120** or **220**, third image capture unit **130** or **230** etc.) of the inspection tool **100** or **200** with a depth of field of substantially 30 μm.

FIG. **5**A shows a schematic front view of an optical fiber core **18***a*, on a V-groove **540** of a specimen **550**, that is misaligned with a waveguide **19** on a silicon PIC.

The specimen **550** of FIG. **5**A may be similar or identical to the specimen **150** of FIG. **1** or the specimen **250** of FIG. **2**A or the specimen **350** of FIG. **3**.

According to an aspect of the present disclosure, the specimen **550** may be a single crystal silicon wafer. At least one V-groove **540** of the specimen **550** may be anisotropically etched out in the single crystal silicon wafer, for example, using one or more photolithographically defined masks (not shown). For example, a width and position of the photolithographically mask(s) (e.g., on the single crystal silicon wafer) as well as an etch rate (e.g., control of an etch rate) may determine a shape and/or position of each formed V-groove **540**. Both wet etching and dry etching may be used in such V-groove **540** formation.

As shown, the optical fiber **18** (i.e. an object) may be guided and supported by opposing sidewalls **541** (e.g., angular surfaces) of the V-groove **540**. During a fiber attachment process, a buffer lid (not shown) may be used to press downwards or against the optical fiber **18** (e.g., in a direction towards a bottom **543** of the V-groove **540**) from the top surface **552** of the specimen **550** to ensure contact (e.g., direct or physical contact) between the optical fiber **18** and the sidewalls **541** of the V-groove **540**. A slope of the sidewalls **541** (e.g., angular surfaces) of the V-groove **540** which may be, for example, about 55 degrees from a horizontal (e.g., 54.74 degrees), may be defined by silicon crystal structure orientation since etching rates in different crystal orientations may be different. Non-ideal shape, location and orientation of the V-groove **540** may lead to misalignment between the optical fiber **18** and the waveguide **19** (as shown in FIG. **5**A, e.g., optical fiber core **18***a* is not aligned with waveguide **19**) in or on the PIC (e.g., silicon PIC), which may result in lower optical coupling efficiency. Any contamination larger than 1 μm on the sidewalls **541** of the V-groove **540** may be another cause for misalignment between the optical fiber **18** and the waveguide **19** since any such contamination (e.g., particle, contaminant etc.) may misposition or misplace or misalign the optical fiber **18** (e.g., with respect to the waveguide **19**) (as shown in FIG. **5**A). Therefore, it may be crucial or important to ensure cleanliness of the V-groove **540** and the optical fiber **18**.

FIG. **5**B is a graph depicting an optical coupling efficiency's dependence on lateral translation misalignment and angular mismatch.

FIG. **5**B may be used to understand a misalignment tolerance between the optical fiber **18** and the waveguide **19** in or on the PIC, with respect to lateral translation and angular degrees of freedom. With reference to FIG. **5**B, a 2 μm of lateral translation misalignment and/or 2 degrees of angular misalignment may lead to about 20% light loss. In aspects involving a photonic integrated device, no more than 20% light loss may be afforded for one single optical interface. Therefore, good alignment both in translation and angular degrees of freedom may be essential.

FIG. **6**A and FIG. **6**B are photographs of prototype inspection tools **601** and **602**, according to various aspects of the present disclosure.

In an aspect, the inspection tools **601** and **602** of FIG. **6**A and FIG. **6**B may be prototypes (e.g., full or partial prototypes) of the inspection tool **100** of FIG. **1** or the inspection tool **200** of FIG. **2**A and FIG. **3**. In an aspect, the prototype inspection tools **601** and **602** may be configured to inspect the specimen **550** of FIG. **5**A.

The prototype inspection tools **601** and **602** may be referred to as an "all-surface V-groove 2D inspection system" and may be capable of detecting any defects and/or any contaminants or particles (e.g., submicron-sized contaminations or particles) on the V-groove **540** of the specimen **550**.

As shown, the prototype inspection tool **601** may include two sidewall orientated vision systems **610** and **620** (i.e. first image capture unit **610** and second image capture unit **620**), which may be similar or identical to the first image capture unit **110** or **210** and the second image capture unit **120** or **220** of the inspection tool **100** of FIG. **1** or the inspection tool **200** of FIG. **2**A and FIG. **3**. Accordingly, the two sidewall orientated vision systems **610** and **620** may be oriented to be directed to the opposing sidewalls **541** (e.g., angular surfaces) of the V-groove **540** of the specimen **550** (e.g., when the specimen **550** is placed on a platform **670**). As an example, in an aspect, each of the two sidewall orientated vision systems **610** and **620** may be or may include a 2-dimensional camera.

In an aspect, the prototype inspection tool **602** may include one top-down vision system **630** (i.e. a third image capture unit **630**), which may be similar or identical to the third image capture unit **130** or **230** of the inspection tool **100** of FIG. **1** or the inspection tool **200** of FIG. **2**A and FIG. **3**. Accordingly, the top-down vision system **630** may be oriented to be directed to a plateau **543** (e.g., horizontal inner surface or floor surface) of the V-groove **540** of the specimen **550** between the opposing sidewalls **541** (e.g., angular surfaces) of the V-groove **540** and/or directed to a top surface **552** of the specimen **550**. As an example, in an aspect, the top-down vision system **630** may be or may include a 2-dimensional camera.

FIG. **7**A show an image captured using the first image capture unit **610** and the second image capture unit **620** of the prototype inspection tool **601**, according to various aspects of the present disclosure.

With reference to FIG. **7**A, the sidewall orientated vision systems **610** and **620** (i.e. image capture units) may be focused on (e.g., only on) the opposing sidewalls **541** (e.g., angular surfaces) of the V-groove **540** (e.g., one vision system/image capture unit to one sidewall) such that the sidewall orientated vision systems **610** and **620** may be capable of capturing a focused or sharp image of the opposing sidewalls **541** but not of the plateau **543** (e.g., horizontal inner surface) and/or the top surface **552** of the specimen **550**.

FIG. **7**B show an image captured using the third image capture unit **630** of the prototype inspection tool **602**, according to various aspects of the present disclosure.

With reference to FIG. **7**B, the top-down vision system **630** may be focused on (e.g., only on) the plateau **543** (e.g., horizontal inner surface) and/or the top surface **552** of the specimen **550** such that the top-down vision system **630** may be capable of capturing a focused or sharp image of the plateau **543** (e.g., horizontal inner surface) and/or the top surface **552** of the specimen **550** but not of the opposing sidewalls **541**.

FIG. **7**C show a composite image **701** formed by stitching the images captured using the first image capture unit **610**, the second image capture unit **620**, and the third image capture unit **630** of the prototype inspection tools **601** and **602**, according to various aspects of the present disclosure.

The images captured by the three imaging systems (i.e. the sidewall orientated vision systems **610** and **620** and the top-down vision system **630**) of the prototype inspection tools **601** and **602** may (e.g., optionally) be stitched together to form a composite image **701** for better visualization of the V-groove(s) (e.g., of a first channel or a secondary channel) of the specimen **550** (e.g., for contamination detection purposes).

Specifically, referring back to FIG. **7**B, the image shown is taken by the top-down vision system **630** configured with a numerical aperture of 0.95 so as to be capable of capturing a focused or sharp image of the plateau **543** (e.g., horizontal inner surface) and/or the top surface **552** of the specimen **550** but not of the opposing sidewalls **541** of a channel. In the image shown in FIG. **7**B, the opposing sidewalls **541** appear darker (e.g., in color and/or contrast) as compared to the plateau **543** (e.g., horizontal inner surface) and/or the top surface **552** of the specimen **550** due to the opposing sidewalls **541** being inclined (e.g., being a steep slope) with respect to an optical axis of the top-down vision system **630**, thereby leading to much lower light collection efficiency by the top-down vision system **630** of these inclined opposing sidewalls **541**.

On the other hand, FIG. **7**A shows an image taken by the sidewall orientated vision systems **610** and **620** configured with a much lower numerical aperture than the top-down vision system **630**, such that the sidewall orientated vision systems **610** and **620** may be capable of capturing a focused or sharp image of the opposing sidewalls **541** of a channel, but not of the plateau **543** (e.g., horizontal inner surface) and/or the top surface **552** of the specimen **550**, since only each sidewall **541** may be perpendicular (e.g., substantially perpendicular) to an optical axis of a corresponding (e.g., one) vision system (e.g., image capture unit) of the sidewall orientated vision systems **610** and **620** but may be inclined or non-perpendicular with respect to the optical axis of the top-down vision system **630**. In an aspect, a depth of field of each vision system (e.g., image capture unit) of the sidewall orientated vision systems **610** and **620** may be configured in a manner such that a pair (e.g., only a pair) of opposing sidewalls **541** (e.g., of a first channel) may appear in focus while other sidewalls **546** (e.g., of one or more secondary channels which may be spaced apart from the first channel by about 200 μm) may be out of focus.

The in-focus sidewall images, e.g., captured by the sidewall orientated vision systems **610** and **620**, of each channel may be cropped and stitched together as shown in FIG. **7**C. In FIG. **7**C, a foreign material **50** (e.g., contaminant) of about 15 μm in diameter is circled. In addition, from the sidewall images shown in FIG. **7**C, it may be readily observed that etched surface(s) **541** and/or **546** of the V-groove **540** of the specimen **550** may not be completely smooth, which may be validated to be around 500 nm by utilizing 3D profilers (not shown).

FIG. **8** is a photograph of another prototype inspection tool **801**, according to various aspects of the present disclosure.

In an aspect, the inspection tool **801** of FIG. **8** may be another prototype (e.g., full or partial prototype) of the inspection tool **100** of FIG. **1** or the inspection tool **200** of FIG. **2**A and FIG. **3**. In an aspect, the prototype inspection tool **801** may be configured to inspect the specimen **550** of FIG. **5**A.

As shown, the prototype inspection tool **801** may include two triangulation sensors (e.g., two 3-dimensional profilers) (i.e. first image capture unit **810** and second image capture unit **820**), which may be similar or identical to the first image capture unit **110** or **210** and the second image capture unit **120** or **220** of the inspection tool **100** of FIG. **1** or the inspection tool **200** of FIG. **2**A and FIG. **3**. Accordingly, the two triangulation sensors **810** and **820** (i.e. two image capture units) may be oriented to be directed to the opposing sidewalls **541** (e.g., two substantially opposing angular surfaces) of the V-groove **540** of the specimen **550** (e.g., when the specimen **550** is placed on a platform **870**).

With the prototype inspection tools **601** and **602** of FIG. **6**A and FIG. **6**B, which may (as an example) use or include 2-dimensional cameras (i.e. 2-dimensional image capture units) for 2-dimensional inspection of a specimen, spatial relation between the various or multiple 2-dimensional cameras (i.e. 2-dimensional image capture units) may not be required or determined (e.g., for an inspection of a specimen). On the other hand, with the prototype inspection tool **801** of FIG. **8**, which may (as an example) use or include 3-dimensional profilers (i.e. 3-dimensional image capture units), spatial relations of or between the 3-dimensional profilers (i.e. image capture units) may be required to be determined or well-defined (e.g., for an inspection of a specimen). Hence, according to various aspects of the present disclosure, use of 3-dimensional profilers (i.e. 3-dimensional image capture units) may involve calibration (e.g., before an inspection of a specimen) to register relative position(s) and/or direction(s) and/or orientation(s) (e.g., with sub-micrometer precision or tolerances) of such 3-dimensional profilers of an inspection tool according to various aspects of the present disclosure. Accordingly, based on the calibrated relative position and/or direction and/or orientation of each 3-dimensional profiler of an inspection tool, according to various aspects of the present disclosure, with respect to each other (e.g., each or all remaining) 3-dimensional profiler(s) of the inspection tool (e.g., prototype inspection tool **801**), a 3-dimensional profile (e.g. line profile) of a surface (e.g., of a V-groove) of a specimen may be produced or obtained via the 3-dimensional profilers of the inspection tool (e.g., by stitching or synthesizing the captured images by the 3-dimensional profilers of the inspection tool together based on data of the relative position and/or direction and/or orientation of each 3-dimensional profiler with respect to each other 3-dimensional profiler of the inspection tool used for inspection of the specimen).

FIG. **9** is a graph **900**, depicting a line profile of V-grooves of a specimen (not shown), obtained using at least two laser confocal 3-dimensional profiler (i.e. image capture units) of an inspection tool (not shown).

FIG. **10** is a table **1000** depicting a magnitude of any optical fiber core misalignment with respect to a waveguide in a PIC (not shown) based on an inspection method using 3-dimensional profilers (i.e. image capture units) of an inspection tool (not shown).

For example, with reference to FIG. **10**, if an optical fiber (e.g., for a semiconductor application) is placed on or in a first V-groove **1001** of a specimen (not shown), it may be observed from the table **1000** that the optical fiber may be vertically misaligned from the waveguide in PIC by 1.39 μm, horizontally misaligned from the waveguide in PIC by 0.25 μm, and radially misaligned from the waveguide in PIC by 1.41 μm.

To more readily understand and put into practical effect the present metrology system and methods for their use in gap measurements, they will now be described by way of examples. For the sake of brevity, duplicate descriptions of features and properties may be omitted.

EXAMPLES

Example 1 provides an inspection method which may include providing a specimen for inspection, the specimen may include a base surface, a top surface opposite the base surface, a first channel positioned in the top surface in a first direction for receiving an object therein. The first channel may be formed by at least two angular surfaces of the specimen, the at least two angular surfaces declining from the top surface of the specimen. The inspection method may further include providing an inspection tool to inspect the specimen, the inspection tool may include (i) a first image capture unit oriented to face a first angular surface of the at least two angular surfaces and (ii) a second image capture unit oriented to face a second angular surface of the at least two angular surfaces. The inspection method may further include capturing a first image of the first angular surface using the first image capture unit and capturing a second image of the second angular surface using the second image capture unit, for inspection of the specimen.

Example 2 may include the method of example 1 and/or any other example disclosed herein, for which an optical axis of the first image capture unit may be substantially perpendicular to the first angular surface and an optical axis of the second image capture unit may be substantially perpendicular to the second angular surface.

Example 3 may include the method of example 1 and/or any other example disclosed herein, for which the first channel further may include a floor surface that is substantially parallel with the base surface of the specimen, for which the floor surface may be adjoined to and may be positioned between the first angular surface and the second angular surface. Further, in Example 3, the inspection tool may further include a third image capture unit oriented to face the floor surface, and the method may include capturing a third image of the floor surface using the third image capture unit.

Example 4 may include the method of example 3 and/or any other example disclosed herein, for which the method may further include stitching the captured first image, second image and third image together to form a composite image of the first channel, and analysing the captured first image, second image and third image using a processor for identifying defects or contamination along the first channel.

Example 5 may include the method of example 1 and/or any other example disclosed herein, for which the first image capture unit and the second image capture unit of the inspection tool may be configured to operate simultaneously to capture the first image of the first angular surface and the second image of the second angular surface at a same time.

Example 6 may include the method of example 1 and/or any other example disclosed herein, for which the method may further include moving the specimen, in the first direction, with respect to the first image capture unit and the second image capture unit of the inspection tool.

Example 7 may include the method of example 1 and/or any other example disclosed herein, for which the specimen may further include a secondary channel, positioned in the top surface of the specimen and in the first direction, for receiving the object therein. Further, in Example 7, the secondary channel may be formed by at least two secondary angular surfaces of the specimen, the at least two secondary angular surfaces declining from the top surface of the specimen. Further, in Example 7, the secondary channel may be adjacent to the first channel.

Example 8 may include the method of example 7 and/or any other example disclosed herein, for which the method may further include moving the specimen, in a second direction substantially perpendicular to the first direction, such that the first image capture unit of the inspection tool faces a first secondary angular surface of the secondary channel and the second image capture unit of the inspection tool faces a second secondary angular surface of the secondary channel, and thereafter, capturing an image of the first secondary angular surface using the first image capture unit and capturing an image of the second secondary angular surface using the second image capture unit.

Example 9 may include the method of example 1 and/or any other example disclosed herein, for which the specimen is a semiconductor substrate and the object is an optical fiber.

Example 10 may include the method of example 1 and/or any other example disclosed herein, for which the method may further include placing the object in the first channel after capturing the first image and the second image.

Example 11 may include the method of example 1 and/or any other example disclosed herein, for which the first angular surface and the second angular surface of the first channel may respectively form an opposing angle with the base surface of the specimen of between substantially 50 degrees to substantially 60 degrees.

Example 12 may include the method of example 1 and/or any other example disclosed herein, for which the first angular surface and the second angular surface of the first channel may form an angle with respect to each other of between substantially 60 degrees to substantially 80 degree.

Example 13 may include the method of example 1 and/or any other example disclosed herein, for which the method may further include configuring each image capture unit of the inspection tool to have a numerical aperture of less than 0.95.

Example 14 may include the method of example 1 and/or any other example disclosed herein, for which the method may further include configuring each image capture unit of the inspection tool with a pixel resolution of substantially 500 nm, a field of view of substantially 3.5 mm by 5 mm, and a depth of field of substantially 30 μm.

Example 15 may include the method of example 1 and/or any other example disclosed herein, for which an opening on the top surface of the specimen for providing access to the first channel has a lateral width of not more than 200 μm.

Example 16 provides an inspection method which may include providing a specimen for inspection, the specimen including a base surface, a top surface opposite the base surface, a substantially V-shaped groove formed in the top surface of the specimen, for receiving an optical fiber therein, the groove being defined by a floor surface that is substantially parallel with the base surface of the specimen and two angular surfaces extending from the floor surface towards an opening of the groove at the top surface, for which the two angular surfaces are declining from the top surface of the specimen, and each of the two angular surfaces forms an angle of between 50 degrees to 60 degrees with respect to the base surface. Further, Example 16 may include providing an inspection tool comprising (i) a first image capture unit configured to face a first of the two angular surfaces and (ii) a second image capture unit configured to face a second of the two angular surfaces, and capturing an image of the first angular surface using the first image capture unit and capturing an image of the second angular surface using the second image capture unit.

Example 17 may include the method of example 16 and/or any other example disclosed herein, for which an optical axis of the first image capture unit of the inspection tool may be substantially perpendicular to the first of the two angular surfaces and an optical axis of the second image capture unit of the inspection tool may be substantially perpendicular to the second of the two angular surfaces.

Example 18 may include the method of example 16 and/or any other example disclosed herein, for which the first image capture unit and the second image capture unit may be configured identically, with a respective numerical aperture of less than 0.95, a respective pixel resolution of substantially 500 nm, a respective field of view of substantially 3.5 mm by 5 mm, and a respective depth of field of substantially 30 μm.

Example 19 may provide an optical inspection tool which may include at least a first image capture unit and a second image capture unit for inspecting specimens having one or more substantially V-shaped grooves, for which the first image capture unit arranged in a first orientation may be directable towards a first angular surface of a first V-shaped groove of a first specimen, and for which the second image capture unit arranged in a second orientation may be directable towards a second angular surface of the first V-shaped groove of the first specimen. Further, the first image capture unit may be configured to capture images of defects or contamination on the first angular surface and the second image capture unit may be configured to capture images of defects or contamination on the second angular surface.

Example 20 may include the optical inspection tool of example 19 and/or any other example disclosed herein, for which the optical inspection tool may further include a third image capture unit arranged in a third orientation being directable towards a bottom surface of the groove positioned between and adjoining the first angular surface and the second angular surface of the V-shaped groove, for which the third image capture unit may be configured to capture images of defects or contamination on the bottom surface. The optical inspection tool may further include a processor coupled to the first image capture unit, the second image capture unit, and the third image capture unit to analyse the captured images by the first image capture unit, the second image capture unit, and the third image capture unit.

While the disclosure has been particularly shown and described with reference to specific aspects, it should be understood by those skilled in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims. The scope of the disclosure is thus indicated by the appended claims and all changes, which come within the meaning and range of equivalency of the claims, are therefore intended to be embraced.

The invention claimed is:

1. An inspection method comprising:

providing a specimen for inspection, the specimen comprising a base surface, a top surface opposite the base surface, a first channel positioned in the top surface in a first direction for receiving an object therein, wherein the first channel is formed by at least two angular surfaces of the specimen, the at least two angular surfaces declining from the top surface of the specimen;

providing an inspection tool to inspect the specimen, the inspection tool comprising (i) a first image capture unit oriented to face a first angular surface of the at least two angular surfaces and (ii) a second image capture unit oriented to face a second angular surface of the at least two angular surfaces;

capturing a first image of the first angular surface using the first image capture unit and capturing a second image of the second angular surface using the second image capture unit, for inspection of the specimen.

2. The method of claim **1**,

wherein an optical axis of the first image capture unit is substantially perpendicular to the first angular surface; and

wherein an optical axis of the second image capture unit is substantially perpendicular to the second angular surface.

3. The method of claim **1**,

wherein the first channel further comprises a floor surface that is substantially parallel with the base surface of the specimen, wherein the floor surface is adjoined to and is positioned between the first angular surface and the second angular surface;

wherein the inspection tool further comprises a third image capture unit oriented to face the floor surface; and

capturing a third image of the floor surface using the third image capture unit.

4. The method of claim **3** further comprising:

stitching the captured first image, second image and third image together to form a composite image of the first channel; and

analysing the captured first image, second image and third image using a processor for identifying defects or contamination along the first channel.

5. The method of claim **1**,

wherein the first image capture unit and the second image capture unit of the inspection tool are configured to operate simultaneously to capture the first image of the first angular surface and the second image of the second angular surface at a same time.

6. The method of claim **1** further comprising:

moving the specimen, in the first direction, with respect to the first image capture unit and the second image capture unit of the inspection tool.

7. The method of claim 1,

wherein the specimen further comprises a secondary channel, positioned in the top surface of the specimen and in the first direction, for receiving the object therein;

wherein the secondary channel is formed by at least two secondary angular surfaces of the specimen, the at least two secondary angular surfaces declining from the top surface of the specimen;

wherein the secondary channel is adjacent to the first channel.

8. The method of claim 7 further comprising:

moving the specimen, in a second direction substantially perpendicular to the first direction, such that the first image capture unit of the inspection tool faces a first secondary angular surface of the secondary channel and the second image capture unit of the inspection tool faces a second secondary angular surface of the secondary channel, and thereafter,

capturing an image of the first secondary angular surface using the first image capture unit and capturing an image of the second secondary angular surface using the second image capture unit.

9. The method of claim 1,

wherein the specimen is a semiconductor substrate; and wherein the object is an optical fiber.

10. The method of claim 1 further comprising:

placing the object in the first channel after capturing the first image and the second image.

11. The method of claim 1,

wherein each of the first angular surface and the second angular surface of the first channel forms an opposing angle with the base surface of the specimen of between substantially 50 degrees to substantially 60 degrees.

12. The method of claim 1,

wherein the first angular surface and the second angular surface of the first channel form an angle with respect to each other of between substantially 60 degrees to substantially 80 degree.

13. The method of claim 1,

configuring each image capture unit of the inspection tool to have a numerical aperture of less than 0.95.

14. The method of claim 1,

configuring each image capture unit of the inspection tool with a pixel resolution of substantially 500 nm, a field of view of substantially 3.5 mm by 5 mm, and a depth of field of substantially 30 μm.

15. The method of claim 1,

wherein an opening on the top surface of the specimen for providing access to the first channel has a lateral width of not more than 200 μm.

16. An inspection method comprising:

providing a specimen for inspection, the specimen comprising a base surface, a top surface opposite the base surface, a substantially V-shaped groove formed in the top surface of the specimen, for receiving an optical fiber therein, the groove being defined by a floor surface that is substantially parallel with the base surface of the specimen and two angular surfaces extending from the floor surface towards an opening of the groove at the top surface, wherein the two angular surfaces are declining from the top surface of the specimen, and each of the two angular surfaces forms an angle of between 50 degrees to 60 degrees with respect to the base surface;

providing an inspection tool comprising (i) a first image capture unit configured to face a first of the two angular surfaces and (ii) a second image capture unit configured to face a second of the two angular surfaces; and

capturing an image of the first angular surface using the first image capture unit and capturing an image of the second angular surface using the second image capture unit.

17. The method of claim 16,

wherein an optical axis of the first image capture unit of the inspection tool is substantially perpendicular to the first of the two angular surfaces; and

wherein an optical axis of the second image capture unit of the inspection tool is substantially perpendicular to the second of the two angular surfaces.

18. The method of claim 16,

wherein the first image capture unit and the second image capture unit are configured identically, with a respective numerical aperture of less than 0.95, a respective pixel resolution of substantially 500 nm, a respective field of view of substantially 3.5 mm by 5 mm, and a respective depth of field of substantially 30 μm.

19. An optical inspection tool comprising:

at least a first image capture unit and a second image capture unit for inspecting specimens having one or more substantially V-shaped grooves;

wherein the first image capture unit is arranged in a first orientation directed towards a first angular surface of a first V-shaped groove of a first specimen, and wherein the second image capture unit is arranged in a second orientation directed towards a second angular surface of the first V-shaped groove of the first specimen;

wherein the first image capture unit is configured to capture images of defects or contamination on the first angular surface and the second image capture unit is configured to capture images of defects or contamination on the second angular surface.

20. The optical inspection tool of claim 19, further comprising:

a third image capture unit arranged in a third orientation is directable towards a bottom surface of the groove positioned between and adjoining the first angular surface and the second angular surface of the V-shaped groove;

wherein the third image capture unit is configured to capture images of defects or contamination on the bottom surface; and

a processor coupled to the first image capture unit, the second image capture unit, and the third image capture unit to analyse the captured images by the first image capture unit, the second image capture unit, and the third image capture unit.

* * * * *